

BioMove

An Intelligent Electric Wheelchair with Integrated Healthcare Monitoring

Graduation Project

Submitted in Partial Fulfillment of the Requirements for the Bachelor's Degree in
Industrial and Energy Technology

Prepared by

- Adham Mohammed
 - Eslam Atef
 - Youssef Attia
 - Youssef Ezzat
-

Supervised by

Dr. Abdelrahman Sayed

Eng. Soha Magdy

Dr. Ahmed Swidan

Faculty of Industrial And Energy Technology
Helwan International Technological University

Academic Year
2025 – 2026

TABLE OF CONTENTS

ACKNOWLEDGMENT.....	9
---------------------	---

ABSTRACT.....	10
---------------	----

LIST OF ABBREVIATIONS.....	11
----------------------------	----

CHAPTER ONE: INTRODUCTION

1.1 Background.....	12
---------------------	----

1.2 Problem Statement.....	13
----------------------------	----

1.3 Motivation.....	13
---------------------	----

1.4 Project Aim.....	14
----------------------	----

1.5 Project Objectives.....	14
-----------------------------	----

1.6 Scope of the Project.....	14
-------------------------------	----

1.7 Methodology.....	15
----------------------	----

1.8 Project Contributions.....	15
--------------------------------	----

1.9 Report Organization.....	15
------------------------------	----

1.10 Chapter Summary.....	16
---------------------------	----

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction.....	17
-----------------------	----

2.2 Assistive Healthcare Technologies.....	17
--	----

2.3 Evolution of Wheelchair Technology.....	18
2.3.1 Manual Wheelchairs.....	18
2.3.2 Electric Wheelchairs.....	18
2.3.3 Smart Wheelchairs.....	18
2.4 Embedded Systems in Smart Wheelchairs.....	19
2.5 Wireless Communication Technologies.....	19
2.6 Health Monitoring Technologies.....	20
2.7 Safety Systems.....	20
2.8 Mobile Applications in Healthcare.....	21
2.9 Related Work.....	22
2.10 Research Gap.....	24
2.11 Comparison with Existing Smart Wheelchairs.....	24
2.12 Chapter Summary.....	25

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction.....	24
3.2 Overall System Architecture.....	24
3.3 Functional Requirements.....	26
3.4 Non-Functional Requirements.....	26
3.5 System Block Diagram.....	27

3.6 System Workflow.....	28
--------------------------	----

3.7 Data Flow Analysis.....	28
-----------------------------	----

3.8 Design Considerations.....	28
--------------------------------	----

CHAPTER FOUR: MECHANICAL AND HARDWARE SYSTEM DESIGN

4.1 Introduction.....	29
-----------------------	----

4.2 Mechanical System Design.....	30
-----------------------------------	----

4.2.1 Chassis Design.....	30
---------------------------	----

4.2.2 Six-Wheel Configuration.....	30
------------------------------------	----

4.2.3 Weight Distribution.....	31
--------------------------------	----

4.2.4 Load Capacity.....	31
--------------------------	----

4.2.5 Maximum Speed.....	31
--------------------------	----

4.3 Power System.....	32
-----------------------	----

4.3.1 Power Supply Configuration.....	32
---------------------------------------	----

4.3.2 Battery Selection.....	33
------------------------------	----

4.3.3 Power Consumption.....	33
------------------------------	----

4.3.4 Battery Operating Time.....	34
-----------------------------------	----

4.3.5 Battery Monitoring.....	34
-------------------------------	----

4.4 Drive System.....	35
-----------------------	----

4.4.1 Motor Selection.....	35
----------------------------	----

4.4.2 Motor Specifications.....	36
4.4.3 Differential Drive Mechanism.....	36
4.4.4 Speed Control.....	37
4.4.5 Performance.....	37
4.5 ESP32 Main Controller.....	38
4.5.1 Why ESP32?.....	38
4.5.2 Comparison with Other Controllers.....	39
4.5.3 Role of ESP32 in BioMove.....	39
4.5.4 Communication Interfaces.....	40
4.5.5 Engineering Advantages.....	40
4.6 Motor Driver (BTS7960).....	41
4.6.1 Why BTS7960?.....	41
4.6.2 Comparison with Other Drivers.....	42
4.6.3 Motor Control Principle.....	42
4.6.4 Differential Drive Control.....	43
4.6.5 Safety During Motor Operation.....	43
4.7 Sensors and Monitoring System.....	44
4.7.1 MAX30102 Pulse Oximeter Sensor.....	45
4.7.2 MLX90614 Infrared Temperature Sensor.....	46

4.7.3 MPU6050 Motion Sensor.....	48
4.7.4 EMG Muscle Sensor.....	53
4.7.5 Ultrasonic Sensor.....	55
4.7.6 NEO-6M GPS Module.....	57
4.8 Hardware Architecture.....	60
4.9 Hardware Integration.....	61

CHAPTER FIVE: EMBEDDED SOFTWARE DESIGN

5.1 Introduction.....	63
5.2 Software Architecture.....	63
5.3 System Initialization.....	64
5.3.1 Serial Communication Initialization.....	65
5.3.2 GPIO Configuration.....	65
5.3.3 Sensor Initialization.....	65
5.3.4 Bluetooth Low Energy Initialization.....	66
5.3.5 PWM Configuration.....	66
5.3.6 FreeRTOS Task Creation.....	66
5.3.7 Mutex Initialization.....	67
5.3.8 Entering Normal Operation.....	67
5.4 Main Control Loop.....	67

5.4.1 Reading Sensor Data.....	68
5.4.2 Safety Evaluation.....	68
5.4.3 Bluetooth Communication.....	69
5.4.4 Processing User Commands.....	69
5.4.5 Motor Control.....	69
5.4.6 Continuous System Monitoring.....	70
5.5 Bluetooth Low Energy Communication.....	71
5.5.1 BLE Server Architecture.....	71
5.5.2 Receiving User Commands.....	71
5.5.3 Sending Sensor Data.....	72
5.5.4 Connection Management.....	73
5.5.5 Data Packet Structure.....	73
5.6 Custom BLE Communication Protocol.....	74
5.6.1 Packet Structure.....	75
5.6.2 Packet Generation.....	76
5.6.3 Packet Parsing.....	76
5.6.4 Communication Reliability.....	76
5.7 Motor Control Algorithm.....	77
5.7.1 Command Processing.....	78

5.7.2 PWM Speed Control.....	79
5.7.3 Differential Drive Steering.....	79
5.7.4 Safety Priority.....	79
5.7.5 Dual Control System.....	80
5.8 Real-Time Task Management Using FreeRTOS.....	81
5.8.1 Why FreeRTOS?.....	81
5.8.2 Background Medical Processing Task.....	82
5.8.3 Task Synchronization Using Mutex.....	82
5.8.4 Advantages of the Implemented Design.....	82
5.9 Safety Algorithms.....	83
5.9.1 Obstacle Detection Algorithm.....	84
5.9.2 Automatic Emergency Stop.....	84
5.9.3 Fall Detection Algorithm.....	84
5.9.4 Communication Loss Protection.....	85
5.9.5 Battery Monitoring Algorithm.....	85
5.9.6 Temperature Monitoring.....	86
5.10 Software Testing.....	87
5.10.1 Bluetooth Communication Test.....	87
5.10.2 Sensor Verification Test.....	87

5.10.3 Motor Control Test.....	88
5.10.4 Safety Function Test.....	88
5.10.5 Communication Loss Test.....	89
5.10.6 Continuous Operation Test.....	89

CHAPTER SIX: IMPLEMENTATION AND TESTING

6.1 Introduction.....	90
6.2 Prototype Implementation.....	91
6.3 Experimental Setup.....	91

CHAPTER SEVEN: CONCLUSION AND FUTURE WORK

7.1 Conclusion.....	93
7.2 Future Work.....	94
Final Remark.....	94

REFERENCES

ACKNOWLEDGMENT

First and foremost, we would like to express our sincere gratitude to Allah for granting us the strength, patience, and determination to complete this graduation project.

We would like to express our deepest appreciation to our supervisor, **Dr. Abdulrahman Sayed & Eng. Soha Magdy**, for their continuous guidance, valuable advice, constructive feedback, and encouragement throughout the development of this project. Their support played a significant role in helping us overcome many technical challenges and improve the quality of our work.

We are also grateful to the faculty members of the **Department of Mechatronics Engineering** at **Helwan International Technological University (HITU)** for providing us with the knowledge and practical experience that helped us successfully complete this project.

Special thanks go to our families and friends for their endless support, patience, and encouragement during every stage of this journey.

Finally, we would like to thank everyone who contributed, directly or indirectly, to the completion of this project.

Team BioMove

ABSTRACT

Mobility limitations can significantly affect an individual's independence, safety, and quality of life. While conventional electric wheelchairs provide basic mobility, many of them lack intelligent features such as health monitoring, real-time safety systems, and smart mobile integration. This project presents **BioMove**, a smart medical wheelchair designed to improve both mobility and user safety by integrating embedded systems, medical sensors, and a mobile application into a single platform.

The proposed system is built around the **ESP32** microcontroller, which serves as the main processing unit for controlling wheelchair movement, collecting sensor data, and communicating with a mobile application through **Bluetooth Low Energy (BLE)**. The wheelchair can be controlled either through a dedicated Flutter mobile application or a wired joystick, giving the user flexibility in different situations.

To enhance safety, the system includes an ultrasonic sensor for obstacle detection, allowing the wheelchair to stop automatically when an object is detected within a predefined safety distance. An MPU6050 sensor continuously monitors the wheelchair's tilt angle and detects possible fall situations, triggering an emergency notification inside the mobile application. The system also monitors the battery level to warn the user when the remaining charge becomes low.

For healthcare monitoring, the wheelchair integrates a MAX30102 sensor to measure heart rate and blood oxygen saturation (SpO_2), in addition to an MLX90614 infrared sensor for body temperature measurement. A GPS module enables real-time location tracking, allowing the wheelchair's position to be displayed on Google Maps within the mobile application.

In addition to monitoring vital signs such as heart rate, blood oxygen saturation, and body temperature, BioMove integrates an Electromyography (EMG) sensor to monitor muscle activity in real time. This feature enhances the healthcare capabilities of the wheelchair by supporting rehabilitation assessment, muscle fatigue monitoring, and future assistive control applications based on muscle signals.

The mobile application, developed using Flutter, provides an intuitive user interface for wheelchair control, live sensor monitoring, emergency notifications, event logging, speed adjustment, and GPS visualization. Communication between the ESP32 and the application is achieved using a lightweight custom BLE protocol optimized for reliable real-time data transmission.

The developed prototype was designed using a custom-built steel chassis with six wheels, powered by two 24 V DC motors and driven by BTS7960 motor drivers. Performance evaluation demonstrated successful operation under a load of up to **100 kg**, with a maximum speed of approximately **8 km/h**. Based on theoretical calculations, the battery system is expected to provide an operating time of approximately **20 hours** under normal operating conditions.

LIST OF ABBREVIATIONS

Abbreviation	Full Meaning
BLE	Bluetooth Low Energy
BPM	Beats Per Minute
GPS	Global Positioning System
GPIO	General Purpose Input/Output
HR	Heart Rate
I ² C	Inter-Integrated Circuit
IMU	Inertial Measurement Unit
MAX30102	Pulse Oximetry and Heart Rate Sensor
MLX90614	Infrared Temperature Sensor
MPU6050	Motion Processing Unit (Accelerometer and Gyroscope)
MTU	Maximum Transmission Unit
PWM	Pulse Width Modulation
RAM	Random Access Memory
RPM	Revolutions Per Minute
RTOS	Real-Time Operating System
SpO ₂	Peripheral Oxygen Saturation
UART	Universal Asynchronous Receiver/Transmitter

CHAPTER ONE

INTRODUCTION

1.1 Background

Mobility plays a fundamental role in maintaining independence and improving the quality of life for people with physical disabilities and elderly individuals. For many users, a wheelchair is not only a means of transportation but also an essential tool that enables them to perform daily activities and participate in society with greater confidence.

Although electric wheelchairs have significantly improved mobility compared to manual wheelchairs, many commercially available models still focus mainly on movement while offering limited support for user safety, health monitoring, and communication with caregivers. As a result, users may remain exposed to risks such as collisions, falls, undetected health changes, or battery depletion without receiving timely warnings.

Recent advances in embedded systems, wireless communication, and mobile application development have created new opportunities to improve assistive mobility devices. Modern microcontrollers, intelligent sensors, and smartphone applications make it possible to transform a conventional electric wheelchair into a smart healthcare platform capable of monitoring the user's condition while providing safer and more reliable mobility.

Based on this concept, the **BioMove** project was developed as a smart medical wheelchair that combines mobility assistance with real-time health monitoring and intelligent safety features. The system is built around an ESP32 microcontroller and communicates with a Flutter mobile application using Bluetooth Low Energy (BLE). It integrates multiple sensors to monitor the user's health, detect potential hazards, and provide instant notifications when abnormal conditions occur.

In addition to wireless control through the mobile application, the wheelchair can also be operated using a wired joystick, allowing the user to choose the most convenient control method. The system also supports GPS-based location tracking, enabling the wheelchair's position to be displayed on an interactive map within the application.

Rather than focusing solely on transportation, BioMove was designed to provide a safer, smarter, and more connected mobility solution that enhances user independence while reducing potential risks during daily use.



1.2 Problem Statement

People who rely on electric wheelchairs often face challenges that go beyond mobility itself. Many available wheelchairs provide reliable movement but lack integrated systems that continuously monitor the user's health, detect dangerous situations, or communicate important information to the user in real time.

For example, a user may approach an obstacle without receiving sufficient warning, experience an unsafe tilt while moving on ramps, or suffer a fall without any emergency notification being generated. Likewise, changes in vital signs such as heart rate, blood oxygen level, or body temperature usually require separate medical devices instead of being integrated into the wheelchair.

Another limitation is that many traditional electric wheelchairs provide only one control method and offer limited interaction with modern smartphone applications. This reduces flexibility and prevents users from accessing useful features such as live monitoring, location tracking, or intelligent notifications.

These limitations motivated the development of BioMove as an integrated smart medical wheelchair capable of combining movement control, healthcare monitoring, safety systems, and wireless communication within a single platform.

1.3 Motivation

The idea behind BioMove originated from the growing need for a wheelchair that offers more than basic transportation. The project aims to improve the daily experience of wheelchair users by combining modern embedded technologies with practical healthcare features.

The motivation for this project can be summarized as follows:

- Improving the independence of wheelchair users.
- Increasing user safety during daily operation.
- Integrating health monitoring into the wheelchair.
- Providing a simple and intuitive mobile application.
- Applying embedded systems and mechatronics concepts in a real-life healthcare application.
- Developing a practical and affordable prototype using commercially available components.

1.4 Project Aim

The main aim of this project is to design and implement a smart medical wheelchair that improves user mobility, enhances safety, and provides real-time health monitoring through the integration of embedded systems, wireless communication, medical sensors, and a mobile application.

1.5 Project Objectives

The objectives of the BioMove project are:

- Design and manufacture a custom steel wheelchair chassis.
 - Develop a reliable electric drive system using two 24 V DC motors.
 - Implement a dual control system using both a Flutter mobile application and a wired joystick.
 - Build an embedded control system based on the ESP32 microcontroller.
 - Monitor heart rate and blood oxygen saturation using the MAX30102 sensor.
 - Measure body temperature using the MLX90614 infrared sensor.
 - Monitor muscle activity using an EMG sensor to support rehabilitation assessment and future assistive control applications.
 - Detect obstacles using an ultrasonic sensor and stop the wheelchair automatically when necessary.
 - Detect unsafe tilt conditions using the MPU6050 sensor.
 - Display the wheelchair location using a GPS module integrated with Google Maps.
 - Monitor battery level and notify the user when the remaining charge becomes low.
 - Develop a Bluetooth Low Energy communication protocol for real-time data exchange between the wheelchair and the mobile application.
-

1.6 Scope of the Project

This project focuses on the design, implementation, and testing of a prototype smart medical wheelchair capable of providing intelligent mobility assistance and health monitoring.

The project includes the mechanical construction of the wheelchair, the electronic hardware, embedded software development, Bluetooth communication, and a Flutter mobile application for monitoring and control.

Medical sensor readings are intended for monitoring purposes only and should not be considered a replacement for certified medical equipment. Similarly, the estimated battery operating time is based on theoretical calculations under normal operating conditions and may vary depending on practical usage.

1.7 Methodology

The development of BioMove followed a structured engineering approach consisting of the following stages:

1. Defining the project requirements and identifying user needs.
 2. Designing and manufacturing the mechanical chassis.
 3. Selecting suitable electronic components and sensors.
 4. Developing the embedded software for the ESP32.
 5. Designing the Flutter mobile application.
 6. Integrating the hardware and software subsystems.
 7. Testing each subsystem individually.
 8. Performing full system integration and evaluating overall performance.
-

1.8 Project Contributions

The BioMove project introduces an integrated solution that combines several technologies into a single assistive mobility platform. The main contributions include:

- A dual-control wheelchair using both a mobile application and a wired joystick.
 - Integration of health monitoring and safety systems within the wheelchair.
 - A lightweight BLE communication protocol for real-time data exchange.
 - Real-time GPS tracking integrated with Google Maps.
 - Automatic obstacle detection and emergency notifications.
 - A custom-designed steel chassis capable of supporting loads up to 100 kg.
-

1.9 Report Organization

This report is organized into seven chapters.

Chapter One introduces the project and presents its objectives, motivation, and overall scope.

Chapter Two reviews previous work and related technologies in the field of smart wheelchairs and assistive healthcare systems.

Chapter Three presents the system analysis, requirements, and overall architecture.

Chapter Four describes the hardware components and mechanical design.

Chapter Five explains the embedded software architecture, communication protocol, control algorithms, and software implementation of the BioMove system.

Chapter Six presents the implementation of the complete prototype, experimental setup, system integration, and performance testing.

Chapter Seven concludes the project, summarizes the achieved objectives, discusses the main contributions, and suggests possible future improvements.

1.10 Chapter Summary

This chapter introduced the BioMove Smart Medical Wheelchair project and discussed the motivation behind its development, the problem it addresses, the project objectives, scope, methodology, and expected contributions. It also provided an overview of the report structure. The following chapter reviews existing research and technologies related to smart wheelchairs, embedded systems, and healthcare monitoring to establish the foundation for the proposed system.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

Assistive technology has become one of the fastest-growing fields in engineering due to its direct impact on improving the quality of life for people with disabilities and elderly individuals. In recent years, significant progress in embedded systems, wireless communication, sensor technology, and mobile application development has enabled the creation of intelligent mobility devices that offer much more than simple transportation.

Among these devices, smart wheelchairs have attracted considerable attention because they combine mechanical systems, electronic control, healthcare monitoring, and communication technologies within a single platform. Unlike conventional electric wheelchairs, modern smart wheelchairs aim to improve user independence while increasing safety and reducing the need for continuous caregiver assistance.

This chapter reviews the evolution of assistive mobility technologies, the development of smart wheelchairs, the embedded technologies commonly used in these systems, and the current state of research. The chapter concludes by identifying the research gap that motivated the development of the proposed BioMove system.

2.2 Assistive Healthcare Technologies

Assistive healthcare technologies refer to devices and systems designed to help individuals perform daily activities safely and independently despite physical or medical limitations. These technologies range from simple mechanical aids to advanced intelligent systems capable of sensing, processing, and responding to environmental and physiological conditions.

Modern assistive devices increasingly rely on embedded systems, wireless communication, cloud services, and mobile applications to provide continuous support for users. Instead of performing a single task, many recent systems combine mobility assistance with healthcare monitoring, navigation, and emergency response. This evolution has transformed assistive devices from passive tools into interactive smart systems capable of adapting to users' needs.

2.3 Evolution of Wheelchair Technology

The development of wheelchair technology can generally be divided into three generations.

2.3.1 Manual Wheelchairs

Manual wheelchairs represent the earliest and most widely available mobility solution. They are lightweight, inexpensive, and easy to maintain. However, they depend entirely on the physical ability of the user or a caregiver to provide movement.

Although manual wheelchairs remain suitable for many users, they are often not appropriate for individuals with severe mobility limitations or reduced upper-body strength.

2.3.2 Electric Wheelchairs

Electric wheelchairs introduced motorized movement, significantly reducing the physical effort required from users. These systems commonly employ rechargeable batteries, DC motors, and joystick controllers to improve mobility and comfort.

Despite these improvements, many commercially available electric wheelchairs still provide limited intelligent functionality. Most focus on movement while offering only basic safety mechanisms and little or no health monitoring capability.

2.3.3 Smart Wheelchairs

The latest generation integrates electronics, embedded systems, intelligent sensors, and communication technologies into traditional electric wheelchairs.

Smart wheelchairs may include features such as:

- Obstacle detection.
- Automatic braking.
- GPS navigation.
- Wireless communication.
- Health monitoring.
- Environmental sensing.
- Emergency notification systems.

These features improve user independence while reducing potential safety risks during daily operation. Recent reviews also show a growing trend toward integrating AI, sensor fusion, and connected mobile applications into smart wheelchair platforms.

2.4 Embedded Systems in Smart Wheelchairs

Embedded systems are the core of modern smart wheelchair technology. They coordinate communication between sensors, actuators, communication modules, and user interfaces while making real-time decisions based on collected data.

Microcontrollers such as the ESP32 have become popular because they combine high processing performance with integrated wireless communication, making them suitable for compact and energy-efficient medical devices.

In BioMove, the ESP32 acts as the central controller responsible for:

- Receiving user commands.
- Processing sensor readings.
- Controlling motor movement.
- Managing Bluetooth communication.
- Executing safety algorithms.
- Transmitting real-time information to the mobile application.

Its dual-core architecture also allows different software tasks to run simultaneously, improving system responsiveness without requiring additional processing hardware.

2.5 Wireless Communication Technologies

Wireless communication is an essential component of modern assistive mobility systems because it enables continuous interaction between the wheelchair and external devices such as smartphones and monitoring applications.

Several wireless technologies are commonly used:

- Wi-Fi
- Bluetooth Classic
- Bluetooth Low Energy (BLE)

Among these technologies, BLE has become one of the most suitable options for embedded healthcare applications because it offers low power consumption, stable short-range communication, and efficient real-time data exchange.

For this reason, BioMove employs Bluetooth Low Energy to establish communication between the ESP32 controller and the Flutter mobile application. The communication protocol was specifically designed to transmit multiple sensor readings within a compact data packet, reducing communication overhead while maintaining fast updates every second.

2.6 Health Monitoring Technologies

Healthcare monitoring has become one of the defining characteristics of modern smart wheelchairs. Instead of relying on external medical equipment, integrated sensors can continuously monitor the user's physiological condition.

In BioMove, three medical sensors are integrated to continuously monitor the user's physiological condition and muscle activity.

MAX30102

The MAX30102 is a medical-grade optical sensor capable of measuring:

- Heart Rate (HR)
- Blood Oxygen Saturation (SpO₂)

These measurements allow the application to display live physiological data and provide continuous monitoring during wheelchair operation.

MLX90614

The MLX90614 is a non-contact infrared temperature sensor used to measure body temperature quickly without direct physical contact.

When abnormal temperature values are detected, the mobile application immediately alerts the user.

EMG Sensor

The EMG (Electromyography) sensor is used to monitor the electrical activity generated by skeletal muscles during contraction. Unlike conventional vital-sign sensors that measure physiological parameters such as heart rate or body temperature, the EMG sensor provides information about the user's muscle activation level.

In BioMove, the EMG sensor continuously acquires muscle activity signals through surface electrodes attached to the user's forearm or another suitable muscle. The analog signal is processed by the ESP32 and transmitted to the Flutter mobile application via Bluetooth Low Energy (BLE), where the muscle activity level is displayed in real time.

Continuous muscle monitoring can assist healthcare providers in evaluating rehabilitation progress, detecting muscle fatigue, and assessing voluntary muscle contractions. Furthermore, the collected EMG data provides a foundation for future assistive control methods, such as gesture recognition or muscle-controlled wheelchair navigation.

By integrating the EMG sensor with the existing health monitoring system, BioMove expands its medical capabilities beyond traditional vital-sign monitoring, making the wheelchair a more comprehensive healthcare and rehabilitation platform.

2.7 Safety Systems

Safety represents one of the primary design objectives of any smart mobility device.

Several technologies have been proposed to reduce risks during wheelchair operation, including collision avoidance, tilt monitoring, emergency communication, and intelligent braking.

BioMove incorporates multiple safety mechanisms.

Obstacle Detection

An ultrasonic sensor continuously measures the distance between the wheelchair and nearby obstacles.

If the measured distance falls below the predefined safety threshold, the system automatically stops the wheelchair to reduce the risk of collision.

Fall Detection

The MPU6050 inertial measurement sensor continuously monitors acceleration and angular motion.

By analyzing tilt angle and sudden movement, the system can detect dangerous situations and immediately generate an emergency notification inside the mobile application.

Battery Monitoring

Continuous battery monitoring helps prevent unexpected power loss during operation.

Whenever the battery level reaches a critical value, the application warns the user so that appropriate action can be taken before the wheelchair stops.

2.8 Mobile Applications in Healthcare

Smartphone applications have become an essential part of modern healthcare systems due to their accessibility, flexibility, and ability to present information in a user-friendly manner.

Flutter has emerged as one of the most widely adopted cross-platform development frameworks because it allows developers to create a single application that operates efficiently on both Android and iOS devices.

In the BioMove project, the mobile application serves as the primary user interface and provides:

- Wireless wheelchair control.
- Live health monitoring.
- GPS location display.
- Speed adjustment.
- Event logging.
- Emergency notifications.
- Battery status monitoring.



Rather than functioning as a simple remote controller, the application acts as a complete monitoring dashboard for the entire wheelchair system.

The Flutter application also displays the user's muscle activity level acquired from the EMG sensor. The measured signal is updated in real time and presented alongside the heart rate, blood oxygen saturation, body temperature, battery level, and wheelchair status. This allows users and healthcare providers to continuously monitor muscle activation during wheelchair operation and rehabilitation sessions through a single unified interface.

2.9 Related Work

Over the past decade, smart wheelchair systems have received increasing attention from researchers due to the rapid development of embedded systems, wireless communication, and healthcare technologies. Many studies have focused on improving user mobility by integrating sensors, intelligent control algorithms, and mobile applications. However, most existing systems concentrate on a limited number of features rather than providing a comprehensive solution that combines mobility, safety, health monitoring, and real-time communication.

One of the most comprehensive review studies in this field analyzed the evolution of smart wheelchair technologies and highlighted the growing use of sensors, microcontrollers, and autonomous navigation systems. The study concluded that future smart wheelchairs should integrate healthcare monitoring, intelligent safety mechanisms, and user-friendly mobile interfaces instead of focusing only on navigation or movement assistance.

Another study proposed a smart wheelchair capable of detecting obstacles using ultrasonic sensors while allowing users to control the wheelchair wirelessly. Although the system improved user safety during navigation, it did not include any medical monitoring functions or emergency notification mechanisms.

A different research project introduced a wheelchair equipped with heart rate monitoring and wireless communication. The collected physiological data could be transmitted to external devices for observation. However, the proposed system lacked obstacle detection, fall monitoring, and location tracking, limiting its practical use in daily outdoor environments.

Researchers have also investigated wheelchair systems that utilize inertial measurement units (IMUs) to monitor wheelchair orientation and detect dangerous tilt angles. These systems demonstrated good performance in detecting abnormal movement but were generally designed as independent safety modules rather than complete smart wheelchair platforms.

Several recent studies have integrated GPS technology into smart wheelchairs to improve outdoor navigation and allow caregivers to monitor user location. While GPS tracking increased user safety, these systems usually relied on separate applications and did not combine location tracking with health monitoring or real-time wheelchair diagnostics.

With the widespread adoption of smartphones, many researchers have developed mobile applications for wheelchair control and monitoring. Most of these applications provide basic movement commands through Bluetooth communication, while only a limited number support continuous visualization of multiple sensor readings, event logging, or intelligent notification systems.

Based on the reviewed literature, it is clear that each proposed system addresses specific aspects of smart wheelchair technology. Some focus on navigation, others on healthcare monitoring, while a few emphasize communication or safety. However, only a limited number of systems attempt to integrate all these functionalities into a single embedded platform suitable for everyday use.

2.10 Research Gap

The literature review indicates that although considerable progress has been made in the development of smart wheelchair technologies, several important limitations remain. Most existing systems focus on solving only one or two specific challenges, such as navigation, obstacle avoidance, or physiological monitoring, while overlooking the need for a fully integrated assistive healthcare platform.

Many reported systems provide mobility assistance without monitoring the user's vital signs, whereas others include healthcare sensors but lack intelligent safety mechanisms such as automatic obstacle avoidance, fall detection, or emergency alerts. Furthermore, several solutions

rely on multiple independent hardware modules or separate mobile applications, resulting in increased system complexity and reduced usability.

Another significant limitation is the absence of a comprehensive embedded platform capable of combining mobility control, healthcare monitoring, wireless communication, location tracking, battery management, and real-time event logging within a single user interface. Additionally, only a limited number of systems support multiple control methods that improve accessibility and operational flexibility.

To overcome these limitations, BioMove introduces a fully integrated smart medical wheelchair that combines intelligent mobility assistance with continuous healthcare monitoring, active safety mechanisms, Bluetooth Low Energy communication, GPS-based location tracking, battery monitoring, emergency notifications, and a Flutter-based mobile application. The proposed system also supports dual control through both a wired joystick and a wireless mobile application, ensuring greater flexibility, improved safety, and enhanced user independence while maintaining a compact and cost-effective embedded architecture.

2.11 Comparison with Existing Smart Wheelchairs

To better evaluate the proposed BioMove system, a comparison was conducted between BioMove and several common smart wheelchair features reported in previous research. The comparison demonstrates that BioMove integrates multiple healthcare, safety, and mobility functions within a single embedded platform, whereas many existing systems focus on only a limited set of capabilities.

Feature	Conventional Electric Wheelchair	Typical Smart Wheelchair	BioMove
Electric Mobility	✓	✓	✓
Smartphone Control	X	✓	✓
Wired Joystick Control	✓	Some Systems	✓
Heart Rate Monitoring	X	Some Systems	✓
Blood Oxygen (SpO ₂) Monitoring	X	Few Systems	✓
Body Temperature Monitoring	X	Few Systems	✓
Muscle Activity (EMG) Monitoring	X	Rare	✓
Obstacle Detection	X	✓	✓
Fall Detection	X	Some Systems	✓
GPS Tracking	X	Some Systems	✓
Battery Monitoring	Limited	✓	✓

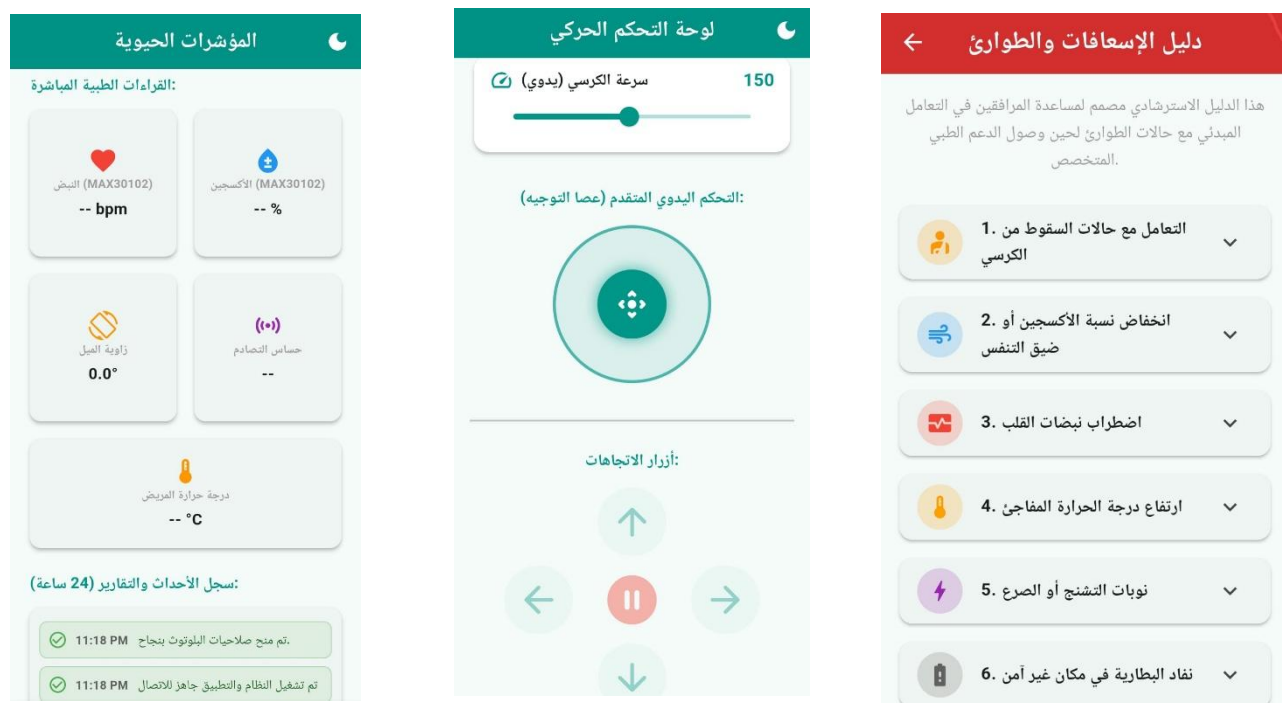
Feature	Conventional Electric Wheelchair	Typical Smart Wheelchair	BioMove
Bluetooth Low Energy (BLE) Communication	X	Some Systems	✓
Flutter Mobile Application	X	Rare	✓
Real-Time Sensor Dashboard	X	Limited	✓
Emergency Notifications	X	Some Systems	✓

The comparison shows that BioMove combines intelligent mobility assistance, healthcare monitoring, rehabilitation support, active safety mechanisms, and wireless communication into a single integrated platform. Unlike many existing smart wheelchair systems that concentrate on navigation or isolated healthcare functions, BioMove provides a comprehensive solution that improves user independence, safety, and continuous health monitoring while maintaining a compact and cost-effective embedded architecture.

2.12 Chapter Summary

This chapter reviewed the evolution of assistive healthcare technologies and smart wheelchairs, highlighting the role of embedded systems, wireless communication, medical sensors, safety mechanisms, and mobile applications in modern mobility solutions.

The review demonstrates that current research increasingly focuses on integrating multiple intelligent features into a single platform. These observations provide the technical foundation for the design of the proposed BioMove Smart Medical Wheelchair, which will be discussed in the following chapter.



CHAPTER THREE

SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

Designing a smart medical wheelchair requires much more than combining electronic components and writing software. The system must operate as a unified platform where mechanical, electrical, embedded, and software subsystems work together reliably and safely.

The development of BioMove followed a system engineering approach in which every subsystem was designed to perform a specific function while maintaining continuous communication with the rest of the system. The ESP32 microcontroller serves as the central processing unit, receiving information from multiple sensors, controlling the wheelchair movement, and exchanging data with the mobile application through Bluetooth Low Energy (BLE).

The main objective of this chapter is to present the complete architecture of the proposed system, describe its functional requirements, explain the interaction between different hardware and software modules, and illustrate how BioMove delivers a safe and intelligent mobility experience.

3.2 Overall System Architecture

BioMove consists of four major subsystems that operate together to achieve the desired functionality.

Mechanical Subsystem

The mechanical subsystem includes the custom-designed steel chassis, six-wheel configuration, two DC drive motors, transmission mechanism, and seating structure. This subsystem provides the physical support required to safely carry the user while maintaining stability during movement.

Embedded Control Subsystem

The embedded subsystem is centered around the ESP32 microcontroller. It is responsible for acquiring sensor readings, processing user commands, controlling the motors, monitoring system status, and communicating with the mobile application.

The ESP32 continuously executes several software tasks to ensure that movement control, sensor monitoring, and wireless communication operate simultaneously without affecting system responsiveness.

Safety and Monitoring Subsystem

The safety subsystem continuously monitors both the user and the surrounding environment.

This subsystem consists of:

- MAX30102 for heart rate and blood oxygen monitoring.
- MLX90614 for body temperature measurement.
- EMG sensor for muscle activity monitoring and rehabilitation assessment.
- MPU6050 for tilt angle and fall detection.
- Ultrasonic sensor for obstacle detection.
- Battery monitoring circuit.
- GPS module for real-time location tracking.

Each sensor contributes to improving user safety by providing continuous feedback to the controller.

Mobile Application Subsystem

The Flutter mobile application acts as the primary user interface.

It allows the user to:

- Control wheelchair movement.
- Adjust motor speed.
- View live sensor readings.
- Monitor battery level.
- Display GPS location on Google Maps.
- Receive emergency notifications.
- Review previous system events through the event logger.

Together, these four subsystems form a complete smart mobility platform capable of providing both transportation and healthcare assistance.

3.3 Functional Requirements

The BioMove system was designed to satisfy several functional requirements based on the daily needs of wheelchair users.

The system shall:

- Move forward.
 - Move backward.
 - Turn left.
 - Turn right.
 - Stop immediately when requested.
 - Stop automatically when an obstacle is detected.
 - Measure heart rate.
 - Measure blood oxygen saturation (SpO₂).
 - Measure body temperature.
 - Monitor muscle activity using the EMG sensor.
 - Monitor wheelchair tilt angle.
 - Detect fall conditions.
 - Determine current GPS location.
 - Display battery level.
 - Adjust movement speed.
 - Exchange data with the mobile application every second.
 - Display all sensor readings in real time.
 - Generate automatic emergency notifications.
 - Store important system events inside the application.
-

3.4 Non-Functional Requirements

In addition to functionality, the system must satisfy several quality requirements.

Reliability

The system should operate continuously while maintaining stable communication between the hardware and the mobile application.

Safety

The wheelchair must automatically respond to dangerous situations whenever possible, such as detecting nearby obstacles.

Real-Time Performance

Sensor readings should be updated every second without noticeable delay.

Ease of Use

The application interface should be simple enough for elderly users and people with limited technical experience.

Maintainability

Hardware components should remain accessible for replacement or maintenance.

Scalability

The system should allow future integration of additional sensors or communication modules without major hardware modifications.



3.5 System Block Diagram

The system begins with user input through either the Flutter mobile application or the wired joystick. Commands are transmitted to the ESP32 controller, which processes the requested action and drives the two DC motors through the BTS7960 motor drivers.

At the same time, the ESP32 continuously collects data from the MAX30102, MLX90614, MPU6050, ultrasonic sensor, GPS module, and battery monitoring circuit.

The processed information is transmitted through Bluetooth Low Energy to the mobile application, where it is displayed as real-time values, graphical indicators, notifications, and map locations.

3.6 System Workflow

The overall operation of BioMove can be summarized as follows:

1. The wheelchair is powered on.
 2. ESP32 initializes all connected sensors.
 3. Bluetooth Low Energy communication is established.
 4. The mobile application connects to the wheelchair.
 5. Sensor readings begin updating every second.
 6. The user sends movement commands using either the application or the wired joystick.
 7. ESP32 processes the commands and controls the motors.
 8. Safety sensors continuously monitor the surrounding environment.
 9. If an obstacle is detected, the wheelchair stops automatically.
 10. If a fall is detected, an emergency notification is generated.
 11. GPS coordinates are updated continuously.
 12. All information is displayed inside the application in real time.
-

3.7 Data Flow Analysis

The BioMove system exchanges information continuously between hardware and software components.

The general data flow follows the sequence below:

Sensors → ESP32 → BLE Communication → Flutter Application → User Interface

Similarly, movement commands follow the reverse direction:

User → Mobile Application → BLE → ESP32 → Motor Driver → DC Motors

This bidirectional communication enables simultaneous monitoring and control while maintaining a responsive user experience.

3.8 Design Considerations

Several engineering factors were considered during the design of BioMove.

Safety was treated as the highest priority. For this reason, automatic obstacle detection and emergency notifications were integrated into the control system.

System simplicity was another important objective. Instead of using multiple controllers, all processing tasks were assigned to a single ESP32 microcontroller, reducing hardware complexity while simplifying software integration.

Energy efficiency was also considered during communication design. Bluetooth Low Energy was selected because it provides reliable wireless communication while consuming significantly less power than many alternative technologies.

Finally, flexibility was improved by supporting two independent control methods: A Flutter mobile application and a wired joystick. This allows the wheelchair to remain operable even if one control method is temporarily unavailable.

CHAPTER FOUR

MECHANICAL AND HARDWARE SYSTEM DESIGN

4.1 Introduction

The hardware design of BioMove represents the foundation upon which all intelligent functions of the wheelchair are built. A reliable smart wheelchair cannot depend solely on advanced software; it also requires a robust mechanical structure, a dependable power system, carefully selected electronic components, and proper integration between all hardware modules.

During the design stage, the primary objective was to develop a wheelchair capable of providing safe movement while supporting continuous health monitoring, environmental sensing, and wireless communication without compromising reliability or user comfort.

Unlike many academic prototypes that rely on preassembled platforms, the mechanical structure of BioMove was designed and fabricated specifically for this project. A custom steel chassis was constructed to support the required electronic components, batteries, motors, and the expected user weight while maintaining structural stability during operation.

The electronic hardware was then integrated around the ESP32 microcontroller, which serves as the central processing unit responsible for coordinating sensor measurements, motor control, communication with the mobile application, and execution of the safety algorithms.

This chapter describes every hardware subsystem used in BioMove, explains its role within the overall system, and discusses the engineering considerations that influenced the final design.

4.2 Mechanical System Design

The mechanical structure is one of the most critical parts of the BioMove smart medical wheelchair. Since the wheelchair is responsible for carrying the user's full weight while supporting all electronic components, the mechanical design had to provide high stability, strength, and reliability under different operating conditions.

Unlike many academic projects that use commercially available wheelchair frames, the BioMove chassis was designed and fabricated specifically for this project. Building the chassis from the ground up allowed the team to determine the dimensions, component locations, and weight distribution according to the project requirements instead of adapting the design to an existing frame.

4.2.1 Chassis Design

The chassis was manufactured using steel sections because of their high mechanical strength, durability, ease of fabrication, and availability in the local market. Since the project was intended to support users weighing up to **100 kg**, structural rigidity was considered a primary design requirement.

The frame was designed to accommodate all major hardware components, including the batteries, motors, ESP32 controller, motor drivers, sensors, and wiring, while maintaining sufficient space for future upgrades if required.

Special attention was given to the placement of heavy components such as the batteries and motors. These components were positioned as low as possible inside the chassis to reduce the center of gravity and improve overall stability during movement.

4.2.2 Six-Wheel Configuration

BioMove employs a six-wheel configuration consisting of two large central driving wheels and four supporting wheels distributed at the front and rear of the chassis.

The central wheels are directly connected to the two 24 V DC motors, providing the driving force required to move the wheelchair. The remaining four wheels act as stabilizing wheels that improve balance and reduce chassis vibration while travelling over uneven surfaces.

Compared with a traditional four-wheel configuration, the six-wheel arrangement provides several engineering advantages.

- Improved mechanical stability.
- Better weight distribution.

- Increased user safety.
- Reduced tipping risk during movement.
- Smoother motion over small obstacles.

Because the driving wheels are located near the center of gravity of the wheelchair, turning requires less effort and provides better maneuverability in confined spaces such as hospitals and indoor environments.

4.2.3 Weight Distribution

Proper weight distribution is essential for maintaining wheelchair stability.

During the design process, heavy components including the batteries and drive motors were installed close to the geometric center of the chassis. This arrangement minimizes uneven loading on the wheels and reduces the possibility of instability during acceleration, braking, or turning.

In addition, placing the batteries near the bottom of the chassis lowers the center of gravity, which improves resistance against tipping and enhances user safety.

4.2.4 Load Capacity

The BioMove prototype was designed and experimentally tested to support users weighing up to **100 kg**.

Mechanical testing demonstrated that the steel chassis maintained its structural integrity under the expected operating load without noticeable deformation. This confirms that the selected materials and structural design are suitable for the intended application.

4.2.5 Maximum Speed

The wheelchair was designed to achieve a maximum speed of approximately **8 km/h**.

This speed represents a practical balance between mobility and user safety. Higher speeds may reduce stability and increase stopping distance, while lower speeds may limit usability during daily transportation.

The selected speed allows comfortable indoor and outdoor movement while maintaining safe operation under normal conditions.

Engineering Insight

The mechanical design of BioMove was not intended to maximize speed or minimize weight. Instead, the design philosophy focused on achieving the highest possible level of stability, reliability, and user safety. Every structural decision, including the selection of steel, the six-wheel configuration, and the placement of heavy components, was made to improve the overall safety of the wheelchair during real-world operation.

Design Challenge

One of the main mechanical challenges was integrating all hardware components within the chassis without affecting user comfort or system stability. Careful planning of component locations ensured balanced weight distribution while leaving sufficient space for maintenance and future modifications.

Practical Note

After assembling the prototype, several adjustments were made to the positions of the batteries and electronic modules before reaching the final layout. These adjustments improved the balance of the wheelchair and reduced unnecessary vibration during movement.

4.3 Power System

The power system is one of the most important subsystems in BioMove because it supplies electrical energy to all hardware components, including the drive motors, sensors, embedded controller, and communication modules. A reliable power source is essential to ensure continuous wheelchair operation and maintain user safety.

4.3.1 Power Supply Configuration

BioMove is powered by two rechargeable **12 V, 12 Ah** batteries connected in series to provide a total output voltage of **24 V**.

This configuration was selected because the drive motors require a 24 V supply to deliver the desired torque and speed. Connecting the batteries in series increases the output voltage while maintaining the battery capacity at **12 Ah**.

The generated 24 V supply is distributed to the different hardware modules according to their operating voltage requirements. Components that require lower voltages are supplied through appropriate voltage regulation circuits to ensure stable and safe operation.

4.3.2 Battery Selection

Several factors were considered during the battery selection process.

The batteries had to provide:

- Sufficient energy for extended operation.
- Stable voltage for motor control.
- Reliable performance under varying loads.
- Easy replacement and maintenance.
- Wide local availability.

A pair of 12 V batteries satisfied these requirements while keeping the electrical system relatively simple.

4.3.3 Power Consumption

The electrical load of BioMove is distributed among several subsystems.

The largest portion of power is consumed by the two DC drive motors during wheelchair movement. The remaining electronic components—including the ESP32 controller, sensors, GPS module, Bluetooth communication, and motor drivers—consume comparatively lower power.

As a result, the total power consumption varies continuously depending on several operating conditions, including:

- User weight.
- Wheelchair speed.
- Ground surface.
- Frequency of acceleration and braking.
- Number of active electronic modules.

For this reason, battery operating time is not constant and changes according to the actual usage conditions.



4.3.4 Battery Operating Time

During practical testing, the wheelchair demonstrated a long operating duration under normal usage conditions before requiring battery recharging.

Since battery discharge depends on motor load and user behavior, the operating time should be considered an approximate value rather than a fixed specification.

This makes the reported operating duration representative of the performed experiments rather than a guaranteed value for all operating conditions.

4.3.5 Battery Monitoring

To improve system reliability, BioMove continuously monitors the battery status and transmits the battery level to the mobile application.

The remaining battery percentage is displayed in real time, allowing the user to monitor the available energy during operation.

Whenever the battery level reaches a predefined low threshold, the application automatically generates a warning notification, encouraging the user to recharge the wheelchair before the battery becomes fully depleted.

This feature helps reduce the possibility of unexpected system shutdown while the wheelchair is in use.

Engineering Insight

Selecting the power system involved balancing operating time, total weight, and system performance. While batteries with larger capacities could increase operating time, they would also increase the overall weight of the wheelchair. Therefore, the selected battery configuration provides a practical compromise between mobility, reliability, and energy availability.

Design Challenge

Integrating the batteries within the steel chassis required careful consideration of weight distribution. Their final position was selected to keep the center of gravity as low as possible, improving stability without reducing the available installation space for the remaining hardware components.

Practical Note

Battery performance was evaluated under practical operating conditions rather than relying solely on theoretical calculations. This approach provided a more realistic estimation of wheelchair operating time and ensured that the selected power system met the project requirements.

4.4 Drive System

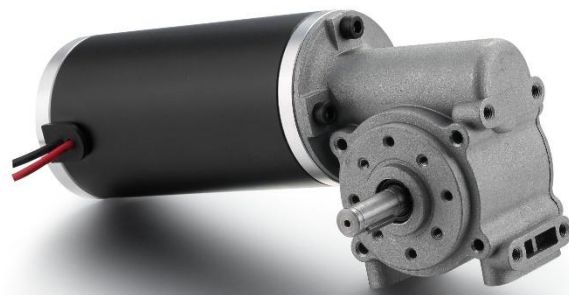
The drive system is responsible for converting electrical energy into mechanical motion, allowing the wheelchair to move safely and smoothly under different operating conditions. Since mobility is the primary function of the system, selecting suitable motors was one of the most important engineering decisions during the hardware design stage.

BioMove employs two **24 V, 250 W DC wheelchair motors**, each dedicated to driving one of the two central wheels. This configuration enables differential drive control, where the speed and direction of each motor can be adjusted independently to perform forward motion, reverse motion, turning, and in-place rotation.

4.4.1 Motor Selection

Several motor options were considered during the planning phase, including standard DC motors and geared motors. However, wheelchair motors were selected because they are specifically designed for mobility applications that require high starting torque, continuous operation, and reliable performance under heavy loads.

Unlike general-purpose DC motors, wheelchair motors are optimized to carry human weight while maintaining stable movement and smooth acceleration. This makes them highly suitable for assistive mobility systems such as BioMove.



4.4.2 Motor Specifications

The wheelchair uses two identical DC motors with the following specifications.

Parameter	Value
Motor Type	Brushed DC Wheelchair Motor
Rated Voltage	24 V
Rated Power	250 W
Number of Motors	2
Drive Method	Differential Drive
Controlled By	BTS7960 Motor Drivers

Each motor provides mechanical power directly to one of the two central driving wheels, allowing the ESP32 controller to regulate the movement of each side independently.

4.4.3 Differential Drive Mechanism

The movement of BioMove is based on the differential drive principle.

Instead of steering the front wheels, the wheelchair changes its direction by varying the speed of the left and right motors.

The movement logic can be summarized as follows:

- Both motors rotate forward at the same speed → Move Forward.
- Both motors rotate backward → Move Backward.
- Left motor slows while the right motor continues → Turn Left.
- Right motor slows while the left motor continues → Turn Right.
- Motors rotate in opposite directions → Rotate in Place.

This steering method provides excellent maneuverability, especially in narrow indoor environments such as hospitals, clinics, and residential buildings.

4.4.4 Speed Control

Motor speed is controlled using Pulse Width Modulation (PWM) signals generated by the ESP32.

Rather than changing the supply voltage, PWM rapidly switches the motor power on and off. By varying the duty cycle of the PWM signal, the average electrical power delivered to the motor changes, resulting in smooth speed adjustment.

The desired speed can be selected through the Flutter mobile application and transmitted to the ESP32 via Bluetooth Low Energy. The controller then updates the PWM duty cycle accordingly, allowing the wheelchair to respond immediately to the user's commands.

4.4.5 Performance

During experimental testing, the completed BioMove prototype successfully transported loads of up to **100 kg** while achieving a maximum speed of approximately **8 km/h**.

These results demonstrate that the selected drive motors provide sufficient torque and power for the intended application while maintaining smooth and stable movement under normal operating conditions.

Engineering Insight

Selecting dedicated wheelchair motors instead of conventional DC motors significantly improved the mechanical reliability of the system. Their high torque characteristics make them suitable for starting from rest while carrying heavy loads, reducing stress on both the motors and the transmission system.

Design Challenge

One of the main challenges during the mechanical design was ensuring that both motors operated synchronously to maintain straight-line motion. Careful PWM control and proper motor alignment were essential to achieve balanced movement and accurate steering.

Practical Note

Practical testing showed that gradual acceleration provided a smoother driving experience and reduced sudden mechanical stress on both the motors and the chassis. This also improved user comfort during normal wheelchair operation.

4.5 ESP32 Main Controller

The ESP32 serves as the central processing unit of the BioMove smart medical wheelchair. It is responsible for coordinating all hardware modules, processing sensor data, controlling the drive motors, and maintaining continuous communication with the mobile application.

Since BioMove integrates multiple sensors, motor control, Bluetooth communication, and safety mechanisms within a single embedded platform, selecting a capable microcontroller was one of the most critical design decisions during the development process.

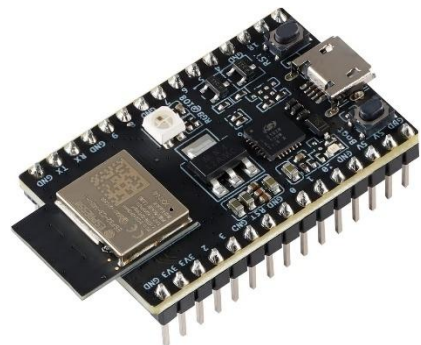
Unlike simple embedded applications that perform only one task at a time, BioMove requires several operations to execute simultaneously. The controller must continuously receive movement commands, monitor medical sensors, detect obstacles, process GPS data, update the battery level, and transmit all information to the Flutter application every second. These requirements made the ESP32 an ideal choice for the project.

4.5.1 Why ESP32?

Several microcontrollers were evaluated during the design phase, including Arduino Uno and Arduino Mega. Although both platforms are widely used in educational projects, they were not suitable for BioMove due to their limited processing capability and lack of built-in wireless communication.

The ESP32 was selected because it combines a powerful dual-core processor, integrated Bluetooth Low Energy (BLE), a large number of GPIO pins, multiple communication interfaces, and low power consumption within a single compact device.

This integration simplified the hardware architecture by eliminating the need for external Bluetooth modules while providing sufficient computational resources for real-time operation.



4.5.2 Comparison with Other Controllers

Feature	Arduino Uno	Arduino Mega	ESP32
Processor	8-bit	8-bit	32-bit Dual-Core
Clock Frequency	16 MHz	16 MHz	Up to 240 MHz
Bluetooth	X	X	✓ Built-in BLE
Wi-Fi	X	X	✓ Built-in
GPIO Pins	Limited	High	High
Processing Performance	Low	Medium	High
Suitable for BioMove	X	Limited	✓

The comparison clearly shows that the ESP32 provides the performance and communication capabilities required for a smart wheelchair that combines mobility control, medical monitoring, and wireless connectivity.

4.5.3 Role of ESP32 in BioMove

Within the BioMove system, the ESP32 performs several essential functions simultaneously:

- Reading heart rate and blood oxygen data from the MAX30102.
- Measuring body temperature using the MLX90614 sensor.
- Reading muscle activity signals from the EMG sensor through the built-in Analog-to-Digital Converter (ADC) and processing the acquired data for real-time monitoring.
- Monitoring wheelchair orientation through the MPU6050.
- Detecting nearby obstacles using the ultrasonic sensor.
- Receiving GPS coordinates from the NEO-6M module.
- Controlling the two DC motors through the BTS7960 motor drivers.
- Receiving movement commands from the wired joystick and the Flutter application.
- Transmitting real-time sensor data to the mobile application via Bluetooth Low Energy.
- Monitoring battery level and generating warning notifications when necessary.

Because all these operations are handled by a single controller, efficient task management is essential to maintain smooth and reliable system performance.

4.5.4 Communication Interfaces

The ESP32 communicates with different hardware modules using multiple communication protocols.

- I²C is used to interface with sensors such as the MAX30102, MLX90614, and MPU6050.
- The built-in Analog-to-Digital Converter (ADC) is used to acquire analog muscle activity signals from the EMG sensor.
- UART is used for communication with the NEO-6M GPS module.
- PWM signals are generated to control the speed of the DC motors through the BTS7960 motor drivers.
- Bluetooth Low Energy (BLE) provides wireless communication with the Flutter mobile application.

Using these interfaces allows the ESP32 to communicate efficiently with all system components without requiring additional controllers.

4.5.5 Engineering Advantages

Selecting the ESP32 provided several important engineering advantages:

- High processing capability.
- Integrated Bluetooth Low Energy communication.
- Excellent support for real-time embedded applications.
- Reduced hardware complexity.
- Lower overall system cost.
- Sufficient processing resources for future system expansion.

These advantages make the ESP32 highly suitable for intelligent healthcare and assistive technology applications.

Engineering Insight

Instead of distributing the system across multiple microcontrollers, BioMove relies on a single ESP32 to coordinate all hardware modules. This approach simplifies system integration, reduces communication delays, minimizes hardware cost, and makes maintenance easier while still providing sufficient processing power for all project requirements.

Design Challenge

Integrating several sensors, wireless communication, and motor control within a single embedded controller required careful software organization to ensure that time-critical operations such as motor control remained responsive while sensor readings and Bluetooth communication continued in the background.

Practical Note

During system integration, each hardware module was tested individually before combining all modules into the final system. This incremental approach simplified troubleshooting and reduced integration time by allowing faults to be identified and corrected at an early stage.

4.6 Motor Driver (BTS7960)

The motor driver acts as the interface between the ESP32 microcontroller and the wheelchair motors. Since the ESP32 operates with low-current digital signals while the drive motors require a much higher current, a dedicated motor driver is necessary to amplify the control signals and deliver sufficient electrical power to the motors.

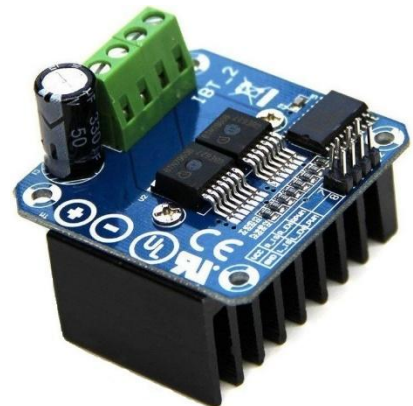
In BioMove, two BTS7960 motor drivers are used, with each driver controlling one 24 V DC wheelchair motor independently. This configuration enables accurate speed regulation and precise steering through differential drive control.

4.6.1 Why BTS7960?

Selecting an appropriate motor driver was a critical step in the hardware design process. The driver had to support high-current operation, provide reliable speed control, and withstand the electrical demands of wheelchair motors.

The BTS7960 was selected because it offers several advantages over commonly used drivers in educational projects.

- High current handling capability.
- Stable operation with 24 V DC motors.
- PWM-based speed control.
- Forward and reverse rotation support.
- Built-in protection against overcurrent and overheating.
- Reliable performance during continuous operation.



These features make the BTS7960 well suited for heavy-duty mobility applications.

4.6.2 Comparison with Other Drivers

Feature	L293D	L298N	BTS7960
Maximum Current	Low	Medium	High
Suitable for 24 V Wheelchair Motors	✗	✗	✓
PWM Speed Control	✓	✓	✓
High Torque Applications	✗	Limited	✓
Recommended for BioMove	✗	✗	✓

Although the L293D and L298N are suitable for small robotic platforms, they cannot reliably drive high-power wheelchair motors. Therefore, the BTS7960 was selected to ensure safe and stable operation.

4.6.3 Motor Control Principle

The ESP32 controls each BTS7960 driver by generating PWM (Pulse Width Modulation) signals.

The duty cycle of the PWM signal determines the average electrical power delivered to the motor.

- Low duty cycle → Low motor speed.
- Medium duty cycle → Moderate motor speed.
- High duty cycle → Maximum motor speed.

Changing the PWM duty cycle allows smooth acceleration and deceleration without altering the supply voltage.

In addition to speed control, the ESP32 controls the rotation direction by selecting the appropriate control inputs of the motor driver.

4.6.4 Differential Drive Control

Since BioMove employs two independent drive motors, each BTS7960 receives separate control signals from the ESP32.

This allows the wheelchair to perform various movements:

Left Motor	Right Motor	Wheelchair Motion
Forward	Forward	Move Forward
Reverse	Reverse	Move Backward
Stop	Stop	Stop
Slow	Fast	Turn Left
Fast	Slow	Turn Right
Forward	Reverse	Rotate in Place

This control strategy provides smooth steering without requiring a mechanical steering mechanism.

4.6.5 Safety During Motor Operation

Safety is an essential consideration in the motor control system.

Whenever the ultrasonic sensor detects an obstacle within the predefined safety distance, the ESP32 immediately stops both motors by disabling the PWM outputs.

Similarly, if the user presses the stop command through the mobile application or the wired joystick, both motor drivers are instructed to stop instantly.

These mechanisms help reduce the risk of collisions and improve user safety during daily operation.

Engineering Insight

The BTS7960 was selected not only because it can supply the required current but also because it provides stable motor operation under varying load conditions. This improves the overall reliability of the wheelchair, especially during acceleration and while carrying heavier users.

Design Challenge

One of the challenges during development was ensuring that both motors responded consistently to the PWM signals. Minor differences between the motors can affect straight-line movement, so careful software calibration was necessary to achieve balanced performance.

Practical Note

Before connecting the wheelchair motors, each BTS7960 driver was individually tested using different PWM values to verify correct direction control, speed regulation, and electrical stability. This testing process simplified the final system integration and reduced troubleshooting time.

4.7 Sensors and Monitoring System

One of the main objectives of BioMove is to provide not only mobility assistance but also continuous health and safety monitoring. To achieve this objective, several sensors were integrated into the system. Each sensor performs a specific task, while the ESP32 combines all collected data and transmits it to the mobile application through Bluetooth Low Energy (BLE).

The selected sensors enable BioMove to monitor the user's health condition, detect dangerous situations, identify surrounding obstacles, and determine the wheelchair's geographical location in real time.

4.7.1 MAX30102 Pulse Oximeter Sensor

The MAX30102 is an optical biomedical sensor designed to measure both heart rate (HR) and blood oxygen saturation (SpO₂). It plays a significant role in BioMove by providing continuous monitoring of two essential vital signs without requiring additional medical equipment.

The sensor operates by emitting red and infrared light into the user's fingertip. Part of this light is absorbed by blood vessels while the remaining light is reflected back to the internal photodetector. Based on the variation in light absorption caused by blood flow, the sensor estimates the heart rate and blood oxygen level.

Within BioMove, the MAX30102 continuously acquires measurements and sends them to the ESP32 through the I²C communication interface. The controller processes the received values and transmits the updated readings to the Flutter application every second, allowing the user to monitor their health status in real time.

Displaying these measurements on the mobile application enables caregivers and users to detect abnormal physiological conditions at an early stage and take appropriate action if necessary.

Why MAX30102?

The MAX30102 was selected because it combines heart rate and SpO₂ measurement within a single compact sensor. It also offers low power consumption, high measurement accuracy, and straightforward integration with the ESP32 through the I²C interface.

Technical Specifications

Parameter	Value
Sensor Type	Optical Pulse Oximeter
Communication	I ² C
Measured Parameters	Heart Rate, SpO ₂
Controller Interface	ESP32
Output	Digital Data



Engineering Insight

Instead of relying on two separate sensors for heart rate and blood oxygen monitoring, the MAX30102 provides both measurements using a single integrated device. This simplifies the hardware design, reduces wiring complexity, and lowers the overall power consumption.

Design Challenge

Accurate measurements require proper finger placement and minimal movement. During testing, it was observed that excessive finger motion could introduce fluctuations in the measured values. Therefore, the application updates the displayed values continuously to provide stable and reliable monitoring.

Practical Note

The displayed heart rate and SpO₂ values are refreshed every second through the Bluetooth connection, allowing users to monitor changes in their physiological condition without noticeable delay.

4.7.2 MLX90614 Infrared Temperature Sensor

Monitoring body temperature is an essential feature in BioMove, as body temperature is one of the primary indicators of a person's health condition. To provide continuous temperature monitoring without causing inconvenience to the user, the MLX90614 infrared temperature sensor was integrated into the system.

Unlike conventional contact temperature sensors, the MLX90614 measures temperature without direct physical contact. It detects the infrared radiation naturally emitted by the human body and converts it into a digital temperature reading. This measurement principle makes the sensor suitable for continuous monitoring in assistive medical devices where user comfort is important.

In BioMove, the sensor continuously measures the user's body temperature and sends the readings to the ESP32 through the I²C communication interface. The controller processes the received data and transmits the updated value to the Flutter application every second.

The measured temperature is displayed on the main dashboard of the application, allowing both the user and caregivers to monitor the user's condition in real time.

High Temperature Warning

To improve user safety, BioMove includes an automatic temperature warning system.

If the measured body temperature exceeds **38.5°C**, the ESP32 immediately sends the updated reading to the mobile application. The application then generates a warning notification informing the user that the measured temperature is above the normal range.

This automatic notification helps users recognize abnormal temperature conditions without continuously monitoring the application.

Why MLX90614?

The MLX90614 was selected for several engineering reasons:

- Non-contact temperature measurement.
- High measurement accuracy.
- Digital I²C communication.
- Fast response time.
- Easy integration with the ESP32.
- Suitable for continuous monitoring applications.

These features make the sensor well suited for smart medical systems such as BioMove.

Technical Specifications

Parameter	Value
Sensor Type	Infrared Temperature Sensor
Communication	I ² C
Measurement Method	Non-contact
Controller Interface	ESP32
Output	Digital Temperature Reading



Engineering Insight

Using a non-contact infrared sensor improves user comfort by eliminating the need for direct skin contact during continuous monitoring. This approach also reduces sensor wear and simplifies daily use of the wheelchair.

Design Challenge

Obtaining stable temperature measurements required selecting an appropriate sensor position that minimizes the influence of surrounding environmental conditions while maintaining a clear view of the user's body.

Practical Note

The application automatically refreshes the displayed temperature every second. Whenever the measured value exceeds the predefined safety threshold, a warning notification is generated without requiring any action from the user.

4.7.3 MPU6050 Motion Sensor

Ensuring user safety is one of the primary objectives of BioMove. Besides monitoring the user's health, the system must also recognize dangerous situations that may occur during wheelchair operation. For this reason, the MPU6050 motion sensor was integrated into the system to monitor the wheelchair's orientation and detect abnormal movements.

The MPU6050 combines a three-axis accelerometer and a three-axis gyroscope within a single integrated circuit. The accelerometer measures linear acceleration along the X, Y, and Z axes, while the gyroscope measures angular velocity around the same axes. By combining data from both sensors, the ESP32 can accurately estimate the orientation and movement of the wheelchair.

In BioMove, the MPU6050 continuously sends motion data to the ESP32 through the I²C communication interface. The controller analyzes these measurements in real time to determine the tilt angle of the wheelchair and identify abnormal situations that may require immediate attention.

Tilt Angle Monitoring

Maintaining wheelchair stability is essential for user safety, especially when moving on ramps or uneven surfaces.

The ESP32 continuously calculates the tilt angle using data received from the MPU6050. The calculated angle is transmitted to the Flutter application every second, allowing the user to monitor the current inclination of the wheelchair in real time.

If the measured tilt angle exceeds the predefined safety limit, the application immediately generates a warning notification, alerting the user that the wheelchair has reached an unsafe inclination.

Continuous tilt monitoring helps reduce the possibility of accidents by informing the user before the wheelchair reaches a dangerous position.

Fall Detection

One of the most important safety features implemented in BioMove is automatic fall detection.

The ESP32 continuously analyzes the acceleration and orientation data received from the MPU6050. When a sudden and abnormal change in motion indicates a possible fall, the controller immediately classifies the event as an emergency.

Once a fall is detected, the system performs the following actions automatically:

- Stops the wheelchair movement.
- Sends a **FALL** emergency message to the mobile application through Bluetooth Low Energy.
- Displays an emergency notification inside the application to inform the user or caregiver.

This immediate response reduces reaction time during emergency situations and improves the overall safety of the wheelchair.

Why MPU6050?

The MPU6050 was selected because it combines both an accelerometer and a gyroscope in a single compact module. This reduces hardware complexity while providing accurate motion sensing for both tilt monitoring and fall detection.

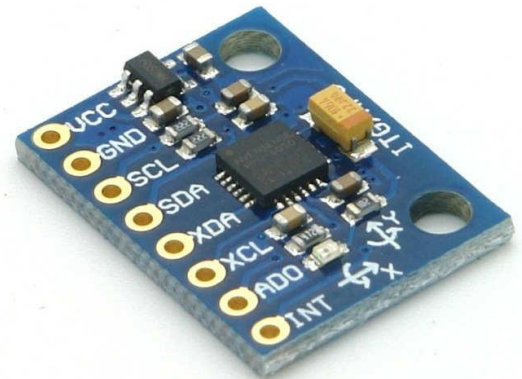
In addition, the sensor offers:

- Digital I²C communication.
- High sampling rate.
- Low power consumption.
- Easy integration with ESP32.
- Stable performance in embedded applications.

These characteristics make it a suitable choice for intelligent mobility systems.

Technical Specifications

Parameter	Value
Sensor Type	6-Axis IMU
Integrated Sensors	3-Axis Accelerometer + 3-Axis Gyroscope
Communication	I ² C
Controller Interface	ESP32
Main Functions	Tilt Monitoring and Fall Detection



Engineering Insight

Instead of using separate sensors for inclination and motion detection, the MPU6050 provides both functions within a single module. This simplifies the hardware design, reduces wiring, and improves synchronization between acceleration and rotational measurements.

Design Challenge

One of the main challenges during development was distinguishing between normal wheelchair movement and an actual fall. Sudden acceleration may occur during rapid turns or while crossing uneven surfaces. Therefore, the software analyzes multiple motion parameters before classifying an event as a fall, reducing the likelihood of false alarms.

Practical Note

The fall detection algorithm was evaluated through repeated practical testing under different movement conditions. The detection thresholds were adjusted several times until the system achieved a reliable balance between fast emergency detection and minimizing false positive alerts during normal wheelchair operation.

4.7.4 EMG Muscle Sensor

Electromyography (EMG) is a biomedical sensing technique used to measure the electrical activity generated by skeletal muscles during contraction. Integrating an EMG sensor into BioMove extends the healthcare capabilities of the wheelchair by enabling continuous monitoring of muscle activity. This feature can assist in evaluating muscle engagement during rehabilitation exercises and may support future developments in assistive control systems.

In the BioMove system, the EMG sensor is attached to the user's forearm or another suitable muscle using surface electrodes. The sensor detects small electrical signals produced by muscle contractions and converts them into analog voltage levels. These signals are acquired by the ESP32 through one of its analog input channels.

The ESP32 continuously samples the EMG signal, performs basic filtering and normalization, and transmits the processed values to the Flutter mobile application via Bluetooth Low Energy (BLE). The application displays the muscle activity level in real time, allowing users or healthcare providers to observe muscle engagement during wheelchair operation or rehabilitation sessions.

Muscle Activity Monitoring

The EMG sensor provides continuous information about the user's muscle activation.

This functionality offers several practical benefits:

- Monitoring muscle contractions during rehabilitation exercises.
- Assessing muscle fatigue over time.
- Evaluating the user's voluntary muscle activity.
- Providing quantitative data for physiotherapy assessment.
- Supporting future assistive control methods based on muscle signals.

Although the current implementation focuses on monitoring muscle activity, the collected EMG data may later be utilized for gesture recognition or muscle-controlled wheelchair navigation.

Why EMG Sensor?

The EMG sensor was selected because it enables non-invasive monitoring of skeletal muscle activity while remaining compact and easy to integrate with embedded systems.

The sensor offers several advantages:

- Real-time muscle activity monitoring.
- Non-invasive surface electrode measurement.
- Simple analog interface.
- Low power consumption.
- Easy integration with the ESP32.
- Suitable for rehabilitation and assistive healthcare applications.

These characteristics make the EMG sensor a valuable addition to the BioMove smart medical wheelchair.

Technical Specifications

Parameter	Value
Sensor Type	Surface EMG Sensor
Communication	Analog Output
Controller Interface	ESP32 ADC
Measured Parameter	Muscle Electrical Activity
Output	Analog Voltage



Engineering Insight

Unlike conventional vital-sign sensors that monitor physiological parameters such as heart rate or body temperature, the EMG sensor provides direct information about the user's muscular activity. This additional layer of monitoring expands the healthcare capabilities of BioMove and supports future intelligent rehabilitation features.

Design Challenge

EMG signals are extremely small and susceptible to electrical noise generated by motors, power electronics, and nearby electronic devices. To improve measurement quality, careful sensor placement, proper electrode attachment, and software filtering were implemented to reduce interference and obtain stable readings.

Practical Note

During experimental testing, the EMG sensor successfully detected voluntary muscle contractions and transmitted the processed activity values to the Flutter application in real time. The displayed readings responded immediately to changes in muscle activation, demonstrating reliable communication between the sensor, ESP32, and the mobile application.

4.7.5 Ultrasonic Sensor

Obstacle detection is one of the most important safety features implemented in BioMove. During wheelchair operation, unexpected obstacles such as walls, furniture, or other objects may appear in the user's path. To reduce the risk of collisions, an ultrasonic distance sensor was integrated into the system.

The ultrasonic sensor continuously measures the distance between the wheelchair and any object located in front of it. It operates by transmitting an ultrasonic pulse and measuring the time required for the reflected echo to return. Based on this time interval, the ESP32 calculates the distance to the detected object.

The measured distance is transmitted to the Flutter application every second, allowing the user to monitor nearby obstacles in real time.

Automatic Obstacle Detection

The ESP32 continuously evaluates the measured distance during wheelchair movement.

When an obstacle is detected within the predefined safety distance, the controller immediately identifies the situation as a potential collision.

Instead of waiting for user intervention, BioMove automatically activates its protection mechanism.

The system immediately performs the following actions:

- Stops both wheelchair motors.
- Prevents further forward movement.
- Sends the updated distance value to the mobile application.
- Displays an automatic warning notification to inform the user.

This autonomous response significantly improves user safety by reducing reaction time and minimizing the possibility of collisions.

Why Ultrasonic Sensor?

The ultrasonic sensor was selected because it provides reliable distance measurements while maintaining a simple hardware design.

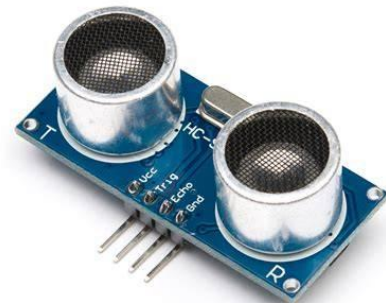
The sensor offers several advantages:

- Non-contact distance measurement.
- Fast response time.
- Low power consumption.
- Simple interface with the ESP32.
- Suitable measuring range for indoor environments.
- Cost-effective solution.

These characteristics make it appropriate for obstacle detection in an assistive mobility system.

Technical Specifications

Parameter	Value
Sensor Type	Ultrasonic Distance Sensor
Measurement Principle	Echo Time-of-Flight
Communication	Digital Trigger/Echo
Controller Interface	ESP32 GPIO
Main Function	Obstacle Detection



Safety Algorithm

Unlike conventional wheelchairs that depend entirely on the user's reaction, BioMove continuously monitors the surrounding environment.

Whenever the measured distance falls below the predefined safety threshold, the ESP32 overrides the movement command and immediately stops both motors before a collision occurs.

This behavior represents an active safety mechanism rather than a passive warning system, making the wheelchair safer for elderly users and individuals with limited reaction capabilities.

Engineering Insight

The ultrasonic sensor was integrated as part of the active safety system rather than as an information-only sensor. Instead of merely displaying the measured distance, the sensor actively participates in controlling wheelchair motion whenever a dangerous situation is detected.

Design Challenge

Selecting an appropriate stopping distance required several practical experiments. A threshold that was too small reduced reaction time, while a very large threshold caused unnecessary interruptions during normal driving. Multiple tests were performed until a suitable balance between safety and usability was achieved.

Practical Note

The obstacle detection system was tested repeatedly using objects of different sizes and distances. During all practical evaluations, the ESP32 successfully interrupted motor operation before contact with the obstacle, demonstrating the effectiveness of the implemented protection algorithm.

4.7.6 NEO-6M GPS Module

One of the key features that distinguishes BioMove from conventional powered wheelchairs is its ability to provide real-time location tracking. This functionality enables caregivers and family members to determine the wheelchair's current position whenever needed, increasing user safety and simplifying assistance during emergencies.

To implement this feature, the BioMove system integrates a NEO-6M GPS module, which receives signals from multiple Global Positioning System (GPS) satellites and calculates the current geographical coordinates of the wheelchair.

The GPS module communicates with the ESP32 through the UART serial interface. Once valid satellite data is received, the ESP32 extracts the latitude and longitude values and includes them in the Bluetooth data packet transmitted to the Flutter application.

The mobile application updates the user's position every second and displays the current location on an interactive Google Maps interface. This allows the wheelchair's movement to be monitored continuously while the map automatically follows the user's position.

Real-Time Location Tracking

The GPS module continuously provides geographical coordinates during wheelchair operation.

Every second, the ESP32 reads the latest GPS data and sends it to the Flutter application using Bluetooth Low Energy (BLE). The application immediately updates the displayed location on the map without requiring any manual refresh.

This continuous synchronization provides an accurate representation of the wheelchair's movement and enables caregivers to monitor the user's location in real time.

Benefits of GPS Integration

Integrating GPS into BioMove provides several practical advantages.

- Continuous location monitoring.
- Improved user safety during outdoor movement.
- Easier assistance in emergency situations.
- Real-time visualization on Google Maps.
- Automatic location updates without user intervention.

These capabilities extend the functionality of the wheelchair beyond mobility assistance by adding an important layer of safety and supervision.

Why NEO-6M?

The NEO-6M module was selected because it provides reliable positioning performance while remaining affordable and easy to integrate with embedded systems.

Additional advantages include:

- Standard UART communication.
- Stable outdoor positioning.
- Compact physical size.
- Low power consumption.
- Wide compatibility with ESP32.

These features make the NEO-6M an appropriate solution for embedded location-tracking applications.

Technical Specifications

Parameter	Value
Module	NEO-6M GPS
Communication	UART
Output Data	Latitude & Longitude
Controller Interface	ESP32
Update Method	Real-Time
Display	Google Maps (Flutter Application)



Engineering Insight

The GPS module was integrated directly into the monitoring system rather than operating as an independent device. This allows location information to be transmitted together with the medical and safety data through a single Bluetooth communication channel, simplifying both software development and system maintenance.

Design Challenge

One challenge encountered during development was ensuring stable location updates while the wheelchair was moving. Since GPS accuracy depends on satellite visibility, the software was designed to continuously update the displayed position whenever new coordinates became available, providing the user with the most recent location information.

Practical Note

The GPS feature was tested by moving the wheelchair through different outdoor locations. During testing, the application successfully updated the wheelchair's position every second and displayed the movement continuously on Google Maps, providing smooth and reliable location tracking.

4.8 Hardware Architecture

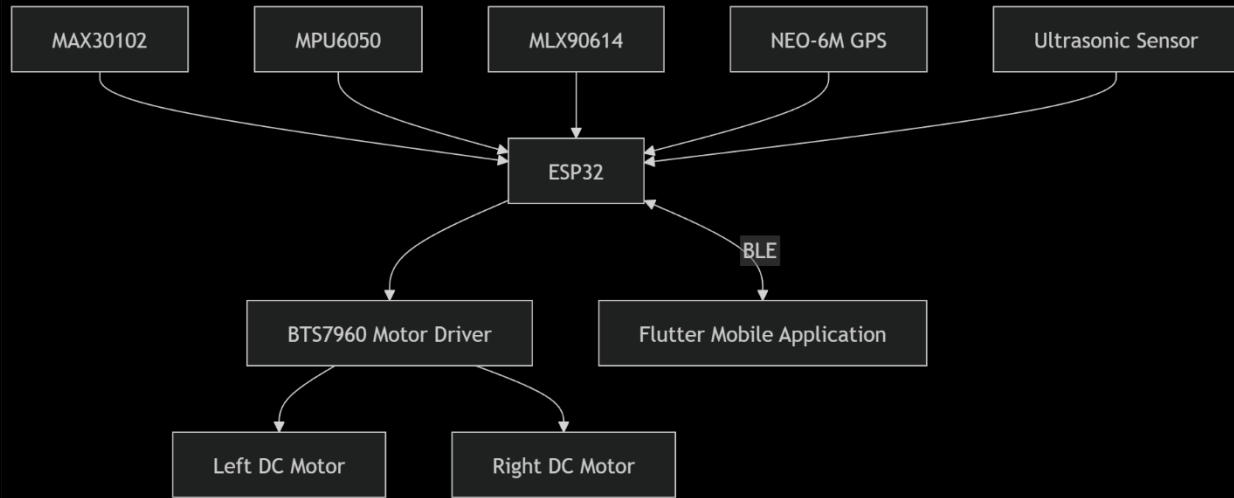
The hardware architecture of BioMove was designed to integrate all electronic modules into a unified embedded system centered around a single ESP32 microcontroller. Rather than using multiple controllers for different subsystems, the project adopts a centralized architecture in which the ESP32 coordinates sensing, motor control, wireless communication, and safety functions simultaneously.

Each hardware module communicates with the ESP32 through the most appropriate communication interface according to its operating requirements. Medical and motion sensors communicate through the I²C bus, the GPS module uses UART serial communication, the ultrasonic sensor is connected through digital GPIO pins, while the BTS7960 motor drivers receive PWM control signals generated by the ESP32.

The ESP32 continuously collects sensor measurements, processes the received data, executes the safety algorithms, and transmits the updated system status to the Flutter mobile application via Bluetooth Low Energy (BLE). At the same time, it receives movement commands from both the mobile application and the wired joystick, ensuring immediate response to user inputs.

This centralized architecture simplifies hardware integration, reduces wiring complexity, minimizes communication latency, and improves system reliability by eliminating the need for multiple interconnected controllers.

Figure 4.8 illustrates the overall hardware architecture of the BioMove system and the relationship between its major hardware components.



BioMove

تجميع المكونات (منظر إيزومتريك)



4.9 Hardware Integration

After completing the mechanical assembly and individual hardware testing, all electronic modules were integrated into a single embedded system centered around the ESP32 microcontroller.

The integration process was carried out gradually to ensure reliable communication between all hardware components before operating the complete wheelchair.

Initially, each sensor was tested individually to verify its functionality and communication with the ESP32. Once individual testing was completed successfully, the sensors were combined into a unified hardware platform using the appropriate communication interfaces.

The MAX30102, MLX90614, and MPU6050 sensors were connected through the shared I²C bus, allowing multiple medical and motion sensors to communicate efficiently using only two signal lines.

The NEO-6M GPS module communicated with the ESP32 through the UART interface, while the ultrasonic sensor used dedicated GPIO pins for trigger and echo signals.

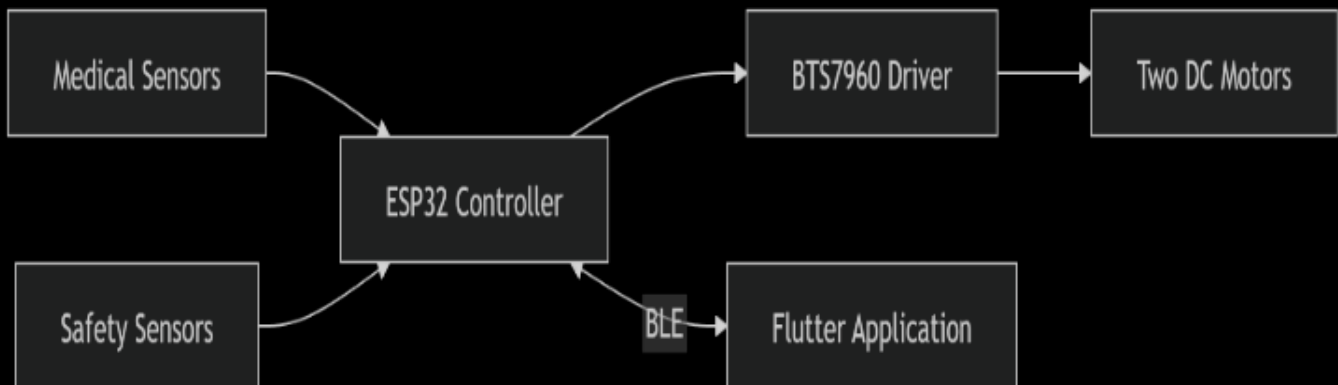
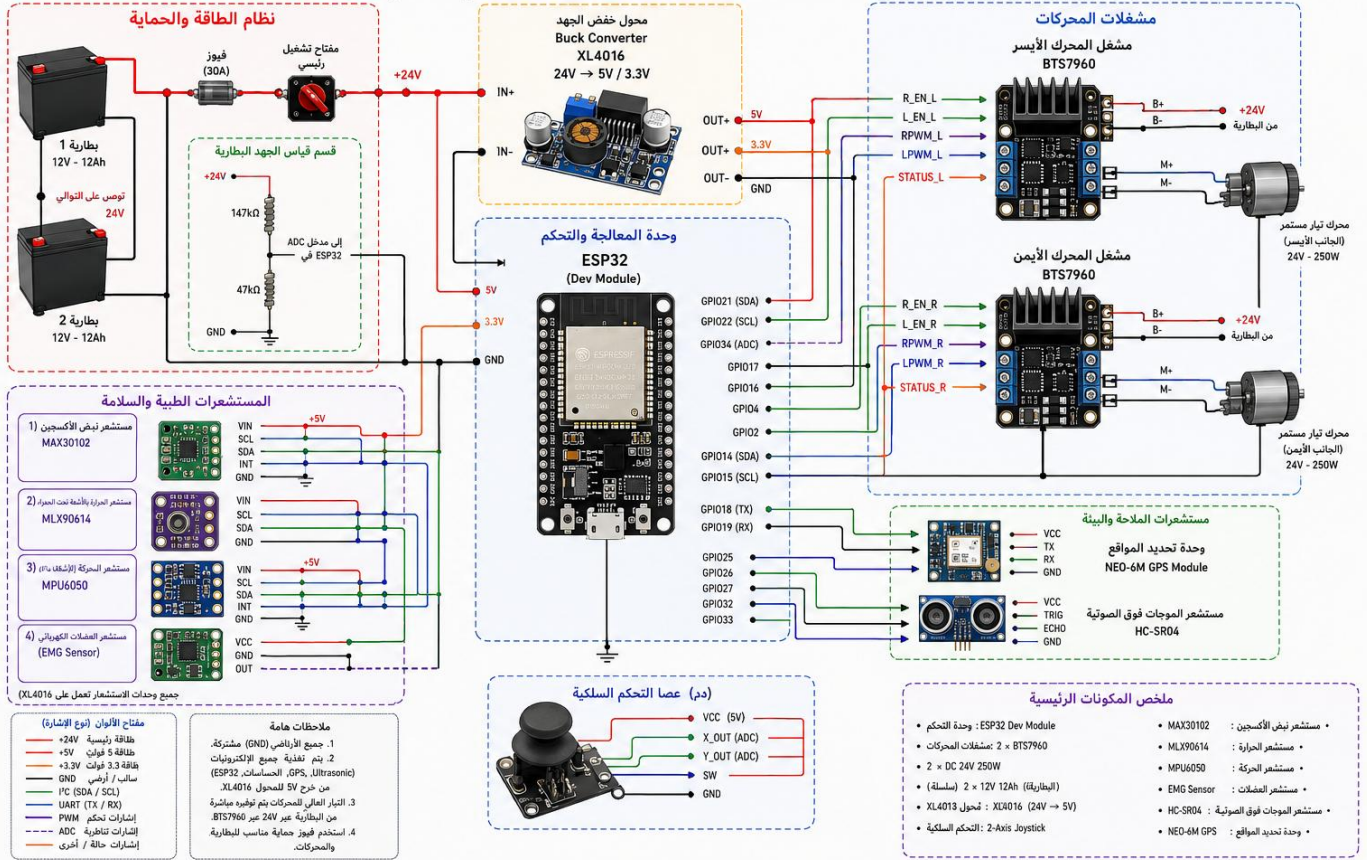
Motor control was implemented independently through two BTS7960 motor drivers connected to separate PWM outputs of the ESP32. This configuration enabled accurate differential drive control for both wheelchair motors.

Bluetooth Low Energy (BLE) established continuous bidirectional communication between the ESP32 and the Flutter mobile application. Sensor readings were transmitted to the application every second, while movement commands and speed settings were received immediately from the user.

The wired joystick was also connected directly to the ESP32, allowing manual control even when the mobile application was unavailable.

The integration of all these subsystems transformed BioMove into a complete intelligent wheelchair capable of monitoring the user's health, ensuring safe navigation, and providing real-time communication through a single embedded platform.

مخطط توصيل نظام BioMove الكرسي الذكي

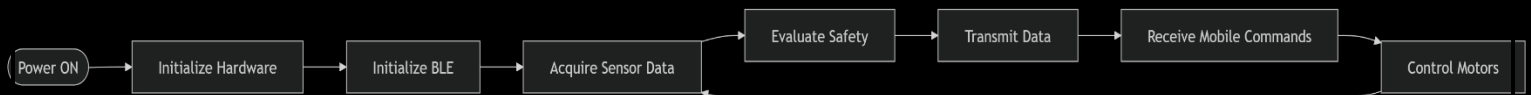


System Operation Sequence

The complete operating sequence of BioMove can be summarized as follows:

1. The ESP32 initializes all hardware modules during system startup.
2. Sensor measurements are collected continuously.
3. The controller processes the received data.
4. Safety algorithms evaluate obstacle distance, wheelchair inclination, and fall conditions.
5. Updated information is transmitted to the Flutter application via Bluetooth Low Energy.
6. The mobile application displays real-time information to the user.
7. Movement commands received from either the application or the wired joystick are executed immediately.
8. If an emergency condition is detected, the ESP32 automatically activates the corresponding protection mechanism.

This workflow ensures continuous monitoring and immediate response throughout wheelchair operation.



Engineering Insight

The success of BioMove does not depend on any single hardware component. Instead, it relies on the seamless integration of multiple embedded subsystems working together under the supervision of the ESP32. This integration transforms separate electronic modules into a unified intelligent assistive system.

Design Challenge

Integrating several communication protocols—including I²C, UART, PWM, GPIO, and Bluetooth Low Energy—within a single embedded controller required careful hardware organization and software scheduling. Proper interface management ensured reliable communication without affecting the responsiveness of the system.

CHAPTER FIVE

Embedded Software Design

5.1 Introduction

The embedded software is the core intelligence of the BioMove smart medical wheelchair. While the hardware components provide sensing, communication, and actuation capabilities, the embedded software coordinates these components and transforms them into a unified intelligent system.

The firmware was developed for the ESP32 microcontroller using the Arduino framework. It is responsible for acquiring sensor data, controlling the wheelchair motors, processing safety conditions, communicating with the Flutter mobile application through Bluetooth Low Energy (BLE), and responding to user commands in real time.

Unlike simple embedded applications that execute a single repetitive task, the BioMove firmware manages multiple operations simultaneously. During normal operation, the controller continuously monitors the user's vital signs, evaluates the surrounding environment, updates the wheelchair location, controls motor movement, supervises battery status, and exchanges data with the mobile application.

To maintain reliable performance, the software was designed with a modular architecture in which each subsystem performs a dedicated function while remaining fully integrated with the rest of the system. This organization improves code readability, simplifies maintenance, and facilitates future expansion.

The objective of this chapter is to present the software architecture of BioMove, explain the interaction between the different software modules, and describe how the embedded firmware enables safe, reliable, and intelligent wheelchair operation.

5.2 Software Architecture

The embedded software of BioMove follows a modular architecture in which each software module is responsible for a specific task. This design approach simplifies debugging, improves readability, and allows new features to be added without significantly modifying the existing code.

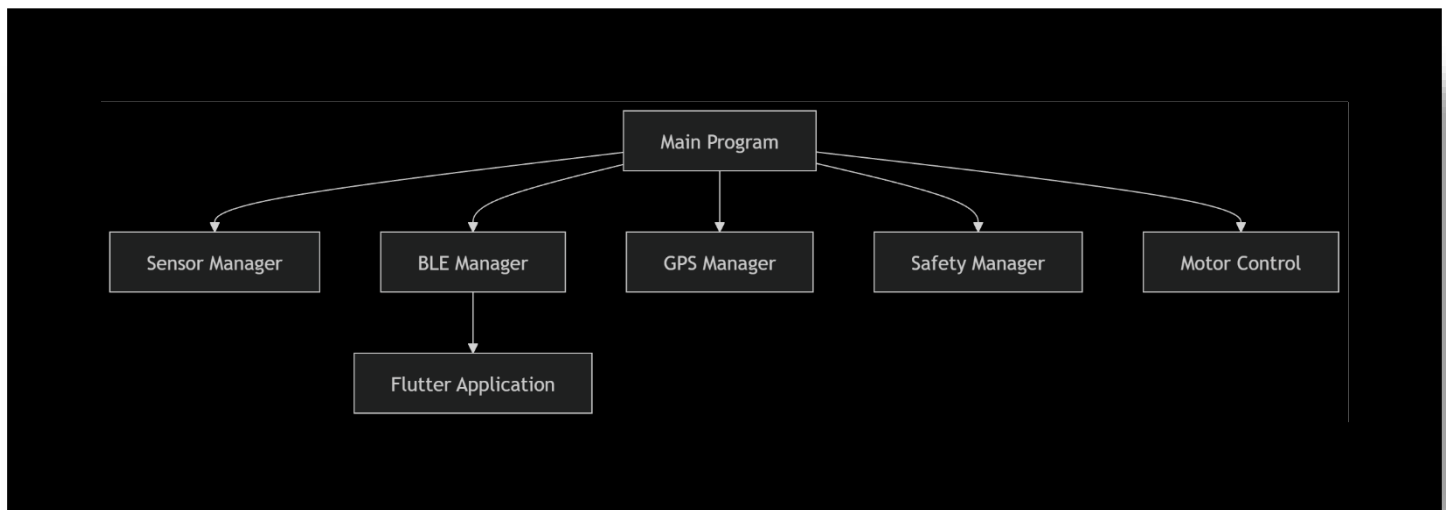
Instead of placing all operations inside a single program loop, the firmware separates sensing, communication, motor control, and safety monitoring into well-defined functional modules.

The main software modules are:

- System Initialization Module
- Sensor Acquisition Module
- Motor Control Module
- Bluetooth Communication Module
- GPS Processing Module
- Safety Monitoring Module
- Battery Monitoring Module
- User Command Processing Module

Each module communicates with the others through shared variables and predefined software routines managed by the ESP32.

This modular organization improves software reliability and ensures that failures in one subsystem can be isolated and diagnosed more easily during maintenance.



5.3 System Initialization

The initialization phase is one of the most important stages of the BioMove firmware. During this stage, the ESP32 prepares all hardware peripherals, communication interfaces, sensors, and software modules before entering normal operation.

All initialization procedures are executed once inside the `setup()` function immediately after the system is powered on. The objective is to ensure that every hardware component is operating correctly before the wheelchair begins accepting user commands.

The initialization process follows a logical sequence to guarantee reliable system startup and prevent communication conflicts between hardware modules.



5.3.1 Serial Communication Initialization

The firmware first initializes the serial communication interface. This interface is mainly used during development and testing to display debugging messages, monitor sensor readings, and verify the execution of different software modules.

Serial monitoring significantly simplified troubleshooting during system integration by allowing the development team to observe the internal behavior of the firmware in real time.

5.3.2 GPIO Configuration

After enabling serial communication, the ESP32 configures all General-Purpose Input/Output (GPIO) pins.

Each pin is assigned according to its specific hardware function, including:

- Motor control signals.
- Ultrasonic trigger output.
- Ultrasonic echo input.
- Battery monitoring input.
- Wired joystick interface.
- PWM motor outputs.

Proper GPIO initialization ensures that every connected device starts in a safe operating state.

5.3.3 Sensor Initialization

Once the hardware pins have been configured, the firmware initializes all sensing devices connected to the ESP32.

The initialization sequence includes:

- MAX30102 heart rate and SpO₂ sensor.
- MLX90614 infrared temperature sensor.
- MPU6050 motion sensor.
- Ultrasonic distance sensor.
- NEO-6M GPS receiver.

Each sensor is verified before normal operation begins. If any device fails during initialization, the corresponding diagnostic message is displayed through the serial monitor to simplify debugging.

5.3.4 Bluetooth Low Energy Initialization

Bluetooth Low Energy (BLE) is initialized during system startup to establish wireless communication with the Flutter mobile application.

The firmware creates the BLE server, registers the communication service and characteristics, and starts advertising automatically. This allows the mobile application to discover and connect to the wheelchair immediately after startup.

Once a connection is established, the ESP32 becomes ready to exchange movement commands and real-time monitoring data.

5.3.5 PWM Configuration

The firmware configures dedicated PWM channels for the two BTS7960 motor drivers.

These channels are responsible for controlling the rotational speed of both wheelchair motors by varying the PWM duty cycle according to commands received from either the mobile application or the wired joystick.

Initializing PWM during startup guarantees stable motor behavior before any movement commands are accepted.

5.3.6 FreeRTOS Task Creation

Unlike conventional Arduino programs that execute all operations inside a single loop, BioMove utilizes the multitasking capabilities of the ESP32 through FreeRTOS.

During initialization, the firmware creates a dedicated background task responsible for processing the MAX30102 sensor and calculating heart rate and blood oxygen saturation values.

Separating this computational task from the main control loop improves system responsiveness and prevents medical data processing from affecting wheelchair control performance.

5.3.7 Mutex Initialization

Since multiple software tasks may access the same I²C sensor simultaneously, the firmware creates a mutex (Mutual Exclusion Semaphore) during initialization.

The mutex guarantees that only one task communicates with the sensor at any given time, preventing communication conflicts and ensuring reliable sensor readings throughout system operation.

5.3.8 Entering Normal Operation

After successfully initializing all peripherals, sensors, communication interfaces, and software tasks, the firmware completes the startup sequence and enters the main control loop.

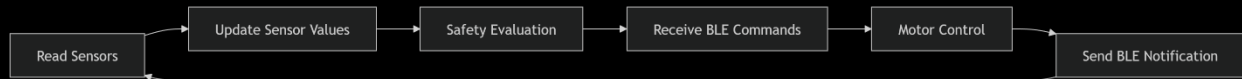
From this point onward, BioMove continuously monitors the user's health, controls wheelchair movement, evaluates safety conditions, exchanges Bluetooth data, and updates the mobile application in real time.

5.4 Main Control Loop

After completing the initialization process, the ESP32 enters the main control loop, which runs continuously throughout the operation of the BioMove wheelchair. The purpose of this loop is to monitor the system, process user commands, execute safety algorithms, and update the mobile application with the latest information.

Unlike traditional embedded applications where all operations are executed sequentially inside a single loop, the BioMove firmware distributes some computational tasks to a dedicated FreeRTOS task. This allows the main loop to remain responsive while medical data processing continues in the background.

During each iteration, the firmware performs a series of operations in a predefined sequence to ensure stable and predictable system behavior.



The first stage of the control loop is acquiring data from the connected sensors.

The ESP32 continuously updates information received from:

- MAX30102 (Heart Rate and SpO₂)
- MLX90614 (Body Temperature)
- MPU6050 (Tilt Angle and Motion)
- Ultrasonic Sensor (Obstacle Distance)
- NEO-6M GPS (Latitude and Longitude)
- Battery Voltage Monitoring

These measurements represent the current state of both the user and the wheelchair.

5.4.2 Safety Evaluation

Immediately after acquiring sensor data, the firmware evaluates the safety status of the system.

Several protection algorithms are executed during every loop iteration, including:

- Obstacle detection.
- Fall detection.
- Tilt monitoring.
- Battery level monitoring.

If any unsafe condition is detected, the ESP32 executes the corresponding safety action without waiting for additional user input.

For example, if an obstacle is detected within the safety threshold, both motors are stopped immediately. Likewise, if a fall is detected, the wheelchair stops automatically and an emergency message is transmitted to the mobile application.

Executing these checks before processing movement commands ensures that user safety always has the highest priority.

5.4.3 Bluetooth Communication

After evaluating system safety, the ESP32 updates the Bluetooth Low Energy connection.

The firmware prepares a data packet containing the latest sensor readings, battery level, GPS coordinates, and wheelchair status. This packet is transmitted to the Flutter application once every second.

At the same time, the firmware continuously checks for incoming Bluetooth commands sent by the user through the application.

This bidirectional communication allows the application to display live information while also controlling wheelchair movement.

5.4.4 Processing User Commands

Movement commands received from either the Flutter application or the wired joystick are interpreted by the ESP32.

The firmware identifies the requested operation, such as moving forward, backward, turning left, turning right, changing speed, or stopping the wheelchair.

Before executing any command, the firmware verifies that no safety condition is currently active. If the system detects an obstacle or another hazardous situation, the requested movement is ignored until safe operation is restored.

This decision-making process prevents unsafe movements while maintaining smooth user control.

5.4.5 Motor Control

After validating the received command, the ESP32 generates the appropriate PWM signals for the two BTS7960 motor drivers.

Each motor is controlled independently, allowing the wheelchair to perform forward movement, reverse movement, turning, or complete stopping.

The duty cycle of the PWM signal determines the rotational speed of each motor, enabling smooth acceleration and precise steering.

5.4.6 Continuous System Monitoring

The firmware continuously repeats the execution cycle throughout the entire operation of the wheelchair.

As new sensor readings become available, the displayed information inside the mobile application is automatically refreshed. At the same time, the ESP32 remains ready to receive new movement commands or react immediately to emergency situations.

This continuous monitoring ensures reliable real-time performance while maintaining a fast response to both user actions and environmental changes.

Engineering Insight

The software design prioritizes safety over mobility. Every movement command is validated against the current safety status before being executed. This guarantees that automatic protection mechanisms always have higher priority than user commands.

Design Challenge

Maintaining smooth motor control while simultaneously processing multiple sensors and Bluetooth communication required careful scheduling of software operations. The use of FreeRTOS helped distribute computational workload and prevented delays that could affect the responsiveness of the wheelchair.

Practical Note

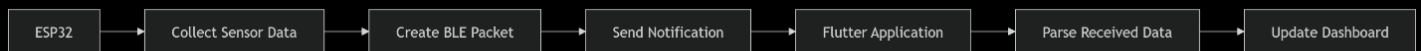
The main control loop was tested under continuous operation during practical experiments. Throughout testing, the firmware maintained stable communication with the mobile application, responded immediately to user commands, and executed safety actions without noticeable delay.

5.5 Bluetooth Low Energy Communication

Wireless communication is one of the fundamental components of the BioMove system. It enables continuous interaction between the ESP32 controller and the Flutter mobile application without requiring physical cables.

To achieve this functionality, Bluetooth Low Energy (BLE) was selected as the primary communication technology due to its low power consumption, fast response, and native support in modern smartphones.

Within the BioMove system, BLE provides bidirectional communication. The mobile application sends movement commands and speed settings to the ESP32, while the controller continuously transmits sensor readings, battery status, GPS coordinates, and safety events back to the application.



5.5.1 BLE Server Architecture

The ESP32 operates as a BLE Server, while the Flutter application functions as a BLE Client.

After powering on, the ESP32 creates a custom BLE service and starts advertising automatically. The mobile application scans for nearby devices and establishes a connection with the wheelchair.

Once connected, both devices begin exchanging data continuously throughout normal operation.

This client-server architecture provides reliable communication while keeping the implementation simple and efficient.

5.5.2 Receiving User Commands

The Flutter application sends movement commands directly to the ESP32 through the BLE write characteristic.

The firmware interprets each received command and immediately performs the requested action.

Supported commands include:

Command	Function
F	Move Forward
B	Move Backward
L	Turn Left
R	Turn Right
S	Stop
Speed Value	Adjust Motor Speed

After validating the command against the current safety conditions, the ESP32 updates the motor control outputs accordingly.

5.5.3 Sending Sensor Data

The ESP32 continuously collects measurements from all connected sensors and prepares a single Bluetooth data packet containing the latest system information.

The transmitted packet includes:

- Heart Rate
- Blood Oxygen Saturation (SpO₂)
- Body Temperature
- Tilt Angle
- Obstacle Distance
- Battery Percentage
- Current Speed
- GPS Coordinates

The firmware sends updated information to the Flutter application once every second, ensuring that the displayed data remains synchronized with the actual state of the wheelchair.

5.5.4 Connection Management

Reliable communication is essential for safe wheelchair operation.

The firmware continuously monitors the BLE connection status.

When a connection is established, normal communication begins immediately.

If the mobile application disconnects unexpectedly, the ESP32 automatically:

- Stops both wheelchair motors.
- Terminates all movement commands.
- Restarts BLE advertising.
- Waits for a new connection.

This protection mechanism prevents uncontrolled wheelchair movement in the event of communication loss.

5.5.5 Data Packet Structure

To minimize communication overhead, BioMove uses a compact custom data format in which all sensor values are transmitted within a single BLE packet.

Each transmitted packet contains multiple labeled fields representing the current system status.

This approach reduces transmission latency, simplifies packet parsing inside the Flutter application, and ensures efficient utilization of the BLE communication channel.

Engineering Insight

Instead of creating separate Bluetooth messages for each sensor, the firmware combines all measurements into a single structured packet before transmission. This design significantly

reduces communication overhead and improves synchronization between the ESP32 and the mobile application.

Design Challenge

One of the challenges encountered during development was maintaining stable Bluetooth communication while simultaneously processing sensor data and motor control tasks. Careful software organization and periodic data transmission ensured reliable communication without affecting system responsiveness.

Practical Note

During experimental testing, the BLE communication maintained stable real-time data exchange throughout wheelchair operation. The mobile application received updated sensor readings every second while movement commands were executed with minimal delay.

5.6 Custom BLE Communication Protocol

To ensure efficient and reliable communication between the ESP32 and the Flutter mobile application, BioMove implements a custom Bluetooth Low Energy (BLE) communication protocol.

Instead of transmitting each sensor reading in a separate Bluetooth message, all system parameters are combined into a single structured data packet. This design minimizes communication overhead, reduces transmission latency, and ensures that all displayed values represent the same system state.

Each transmitted packet contains labeled fields separated by a predefined delimiter. The labels identify the type of data, while the delimiter enables the mobile application to separate and process each value correctly.

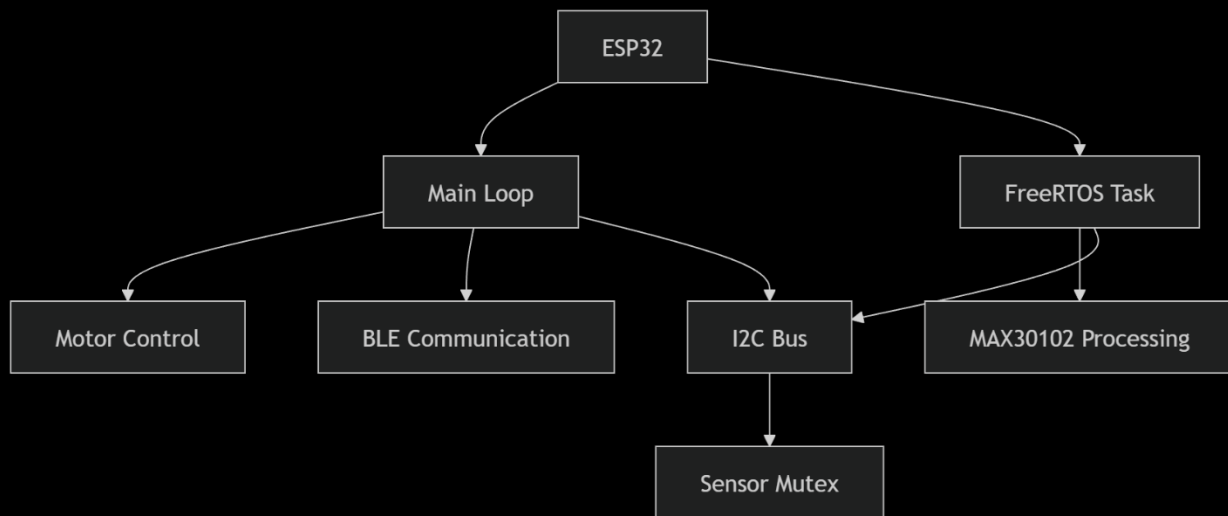
A typical transmitted packet is illustrated below:

```
HR:78|OX:98|TMP:36.7|TLT:4.5|DST:120|BATT:85|SPD:60|GPS:29.8712,31.3125
```

Each field consists of two parts:

- A short identifier (Label).
- The corresponding measured value.

This structure allows both the ESP32 firmware and the Flutter application to exchange multiple measurements through a single BLE notification.



5.6.1 Packet Structure

The communication packet contains all important information required by the mobile application.

Label	Description
HR	Heart Rate (BPM)
OX	Blood Oxygen Saturation (SpO ₂)
TMP	Body Temperature (°C)

Label	Description
TLT	Wheelchair Tilt Angle
DST	Distance to Obstacle
BATT	Battery Percentage
SPD	Current Motor Speed
GPS	Latitude and Longitude

Using descriptive labels makes the protocol easy to understand, maintain, and expand in future software versions.

5.6.2 Packet Generation

The ESP32 periodically reads the latest values from all sensors and combines them into a single formatted string before transmission.

This process ensures that all measurements belong to the same sampling period, improving synchronization between the embedded system and the mobile application.

The complete packet is transmitted once every second through the BLE notification characteristic.

5.6.3 Packet Parsing

After receiving the BLE notification, the Flutter application stores the incoming characters inside an internal communication buffer.

The received packet is then divided into individual fields using the delimiter character (|).

Each field is analyzed separately to determine its corresponding sensor value. The application updates the dashboard immediately after successfully parsing the packet.

This parsing mechanism allows multiple measurements to be processed efficiently while keeping the communication protocol lightweight.

5.6.4 Communication Reliability

Bluetooth Low Energy transmissions may occasionally arrive as fragmented packets because of MTU limitations.

To prevent incomplete data processing, the Flutter application temporarily stores received data inside an internal communication buffer until the complete packet is available.

Only after receiving the full packet does the application begin parsing the sensor values.

This mechanism significantly improves communication reliability and prevents incorrect or incomplete values from appearing on the user interface.

Engineering Insight

Developing a custom communication protocol provided greater flexibility than transmitting individual sensor values separately. The protocol reduced Bluetooth traffic, simplified application development, and ensured that all measurements remained synchronized during real-time operation.

Design Challenge

One of the primary software challenges was handling incomplete BLE packets caused by Bluetooth packet fragmentation. This issue was solved by implementing a buffering mechanism inside the Flutter application, ensuring that only complete packets were processed.

Practical Note

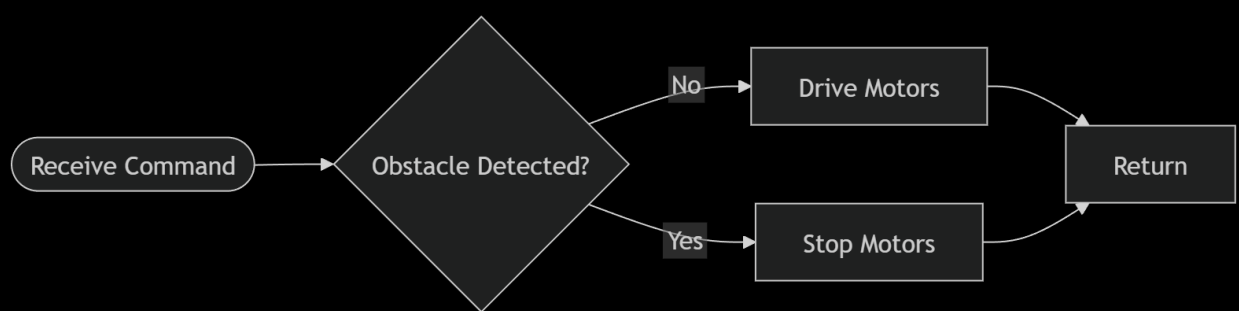
Throughout practical testing, the custom BLE protocol successfully maintained synchronized communication between the ESP32 and the Flutter application. Sensor readings, GPS coordinates, battery information, and movement status were transmitted together with consistent timing and reliable performance.

5.7 Motor Control Algorithm

The motor control algorithm is responsible for converting user commands into physical wheelchair movement while ensuring that all safety conditions are satisfied before any motion is executed.

BioMove uses a differential drive system consisting of two independent 24 V DC wheelchair motors. Each motor is controlled separately through a dedicated BTS7960 motor driver, allowing the wheelchair to move forward, backward, turn, and stop with high accuracy.

Movement commands can be received from either the Flutter mobile application through Bluetooth Low Energy (BLE) or from the wired joystick connected directly to the ESP32. Regardless of the command source, all requests are processed by the same control algorithm to ensure consistent system behavior.



5.7.1 Command Processing

The ESP32 continuously monitors incoming control commands.

Each command represents a specific movement operation:

Command	Action
F	Move Forward
B	Move Backward
L	Turn Left
R	Turn Right
S	Stop Immediately

After receiving a command, the firmware validates it before updating the motor outputs.

If the command is valid and no safety condition is active, the requested movement is executed immediately.

5.7.2 PWM Speed Control

Motor speed is controlled using Pulse Width Modulation (PWM).

Instead of changing the supply voltage, the ESP32 adjusts the PWM duty cycle applied to the BTS7960 motor drivers.

Increasing the duty cycle increases the average power delivered to the motors, resulting in higher wheelchair speed. Likewise, decreasing the duty cycle reduces the motor speed, providing smooth and controllable motion.

The selected speed value is also transmitted to the Flutter application, allowing the user to monitor the current operating speed.

5.7.3 Differential Drive Steering

Steering is achieved by independently controlling the left and right motors.

When both motors rotate at the same speed and direction, the wheelchair moves in a straight line.

To perform a turn, the ESP32 changes the speed of one motor relative to the other. This differential drive technique provides smooth steering without requiring a mechanical steering mechanism.

The control algorithm allows the wheelchair to perform:

- Forward movement.
- Reverse movement.
- Left turn.
- Right turn.
- Complete stop.

This approach improves maneuverability, especially in indoor environments with limited space.

5.7.4 Safety Priority

Although movement commands are received continuously, user safety always has the highest priority.

Before executing any movement command, the firmware verifies the current safety status of the wheelchair.

If one of the following conditions is detected:

- An obstacle within the safety distance.
- A fall events.
- Any critical safety conditions.

The ESP32 immediately overrides the user command and stops both motors.

Only after the unsafe condition has been cleared can normal movement resume.

This priority-based control strategy prevents dangerous situations while maintaining responsive wheelchair operation.

5.7.5 Dual Control System

BioMove supports two independent control methods:

- Flutter mobile application.
- Wired joystick.

Both interfaces communicate with the same motor control algorithm implemented on the ESP32.

This design ensures identical wheelchair behavior regardless of the selected control method and provides an alternative control option if one interface becomes unavailable.

Engineering Insight

The motor control algorithm separates command interpretation from motor actuation. This modular approach allows additional control methods, such as voice commands or autonomous navigation, to be integrated in future versions without redesigning the motor control logic.

Design Challenge

One of the major challenges during firmware development was ensuring smooth transitions between different movement commands while maintaining immediate response to emergency conditions. The final implementation balances responsive control with reliable safety monitoring, resulting in stable wheelchair operation.

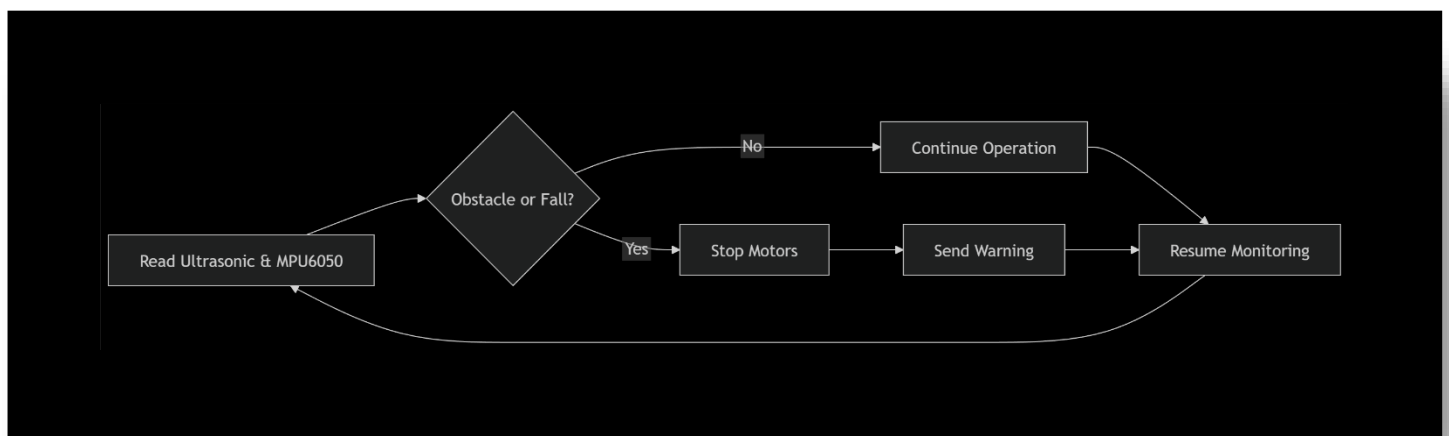
Practical Note

The motor control algorithm was validated through repeated practical testing using different movement patterns, including forward motion, reverse motion, turning, and emergency stopping. During testing, the wheelchair responded consistently to user commands while maintaining immediate reaction to safety events.

5.8 Real-Time Task Management Using FreeRTOS

One of the key software features implemented in BioMove is the use of the FreeRTOS real-time operating system, which is natively supported by the ESP32. Unlike traditional Arduino programs that execute all operations sequentially inside a single loop, FreeRTOS allows multiple tasks to run concurrently while efficiently sharing the processor resources.

The BioMove firmware utilizes FreeRTOS to separate computationally intensive medical signal processing from the main control loop. This design improves system responsiveness and ensures that critical operations such as motor control and safety monitoring remain unaffected by sensor processing.



5.8.1 Why FreeRTOS?

The firmware simultaneously performs several operations, including:

- Reading medical sensors.
- Monitoring obstacle distance.
- Detecting falls.
- Receiving Bluetooth commands.
- Updating GPS coordinates.
- Controlling wheelchair motors.
- Sending real-time data to the mobile application.

Executing all these operations sequentially inside a single loop could reduce system responsiveness, particularly during medical signal processing.

For this reason, FreeRTOS was adopted to distribute the workload into independent tasks.

5.8.2 Background Medical Processing Task

A dedicated FreeRTOS task was created to process the MAX30102 sensor data and calculate the user's heart rate and blood oxygen saturation.

This task executes independently of the main control loop, allowing continuous medical monitoring without interrupting wheelchair control.

While the background task performs signal processing, the main firmware continues handling movement commands, obstacle detection, Bluetooth communication, and other time-critical operations.

This separation significantly improves the real-time performance of the system.

5.8.3 Task Synchronization Using Mutex

Since multiple software modules may access the I²C bus, simultaneous communication with the same sensor could lead to data corruption or communication conflicts.

To avoid this problem, the firmware creates a Mutual Exclusion Semaphore (Mutex).

Before accessing the shared sensor, a task requests ownership of the mutex. Once the communication is completed, the mutex is released, allowing another task to access the sensor.

This synchronization mechanism guarantees reliable sensor communication while preventing race conditions.

5.8.4 Advantages of the Implemented Design

Using FreeRTOS provides several engineering benefits:

- Improved system responsiveness.
- Continuous medical monitoring.
- Independent execution of background tasks.
- Better processor utilization.
- Reliable synchronization between software modules.
- Simplified future software expansion.

These advantages contribute directly to the stability and reliability of the BioMove embedded system.

Engineering Insight

The use of FreeRTOS transforms the firmware from a simple sequential program into a real-time embedded application capable of handling multiple concurrent operations. This architecture allows safety monitoring, medical processing, communication, and motor control to operate together without significantly affecting one another.

Design Challenge

Medical signal processing from the MAX30102 requires continuous calculations that could delay the execution of other software functions if implemented inside the main control loop. Moving these calculations to a dedicated FreeRTOS task eliminated this bottleneck while preserving fast system response.

Practical Note

During experimental testing, the FreeRTOS implementation maintained stable wheelchair operation while continuously updating medical measurements, processing Bluetooth communication, and executing safety algorithms without noticeable performance degradation.

5.9 Safety Algorithms

Safety is one of the primary design objectives of the BioMove smart medical wheelchair. Since the system is intended for elderly users and individuals with mobility impairments, the firmware continuously monitors both the wheelchair and its surrounding environment to reduce the risk of accidents.

Unlike conventional powered wheelchairs that execute user commands directly, the BioMove firmware evaluates every movement request before allowing the wheelchair to move. If a dangerous condition is detected, the firmware automatically overrides the user command and activates the appropriate safety mechanism.

The implemented safety system consists of several independent algorithms working together in real time.

5.9.1 Obstacle Detection Algorithm

The ultrasonic sensor continuously measures the distance between the wheelchair and objects located in front of it.

During each iteration of the main control loop, the measured distance is compared with a predefined safety threshold.

If the detected obstacle is closer than the allowed distance, the ESP32 immediately stops both motors to prevent collision.

At the same time, the updated obstacle distance is transmitted to the Flutter application, allowing the user to monitor nearby objects.

This automatic protection mechanism significantly improves user safety during indoor and outdoor navigation.

5.9.2 Automatic Emergency Stop

The emergency stop function is directly linked to the obstacle detection system.

Whenever an unsafe obstacle is detected, the firmware immediately disables motor movement regardless of the user's requested command.

This automatic intervention occurs within the embedded controller itself and does not depend on the mobile application.

As a result, the wheelchair can react immediately even if wireless communication experiences temporary delays.

5.9.3 Fall Detection Algorithm

The MPU6050 sensor continuously measures the wheelchair orientation and tilt angle.

The firmware analyzes these measurements to detect abnormal inclination that may indicate a fall or a dangerous loss of stability.

When a fall event is detected, the ESP32 performs the following actions automatically:

- Stops both motors.
- Generates a FALL event.
- Sends an emergency notification to the Flutter application.

The application immediately displays a warning message to inform the caregiver or user that a dangerous situation has occurred.

5.9.4 Communication Loss Protection

Reliable wireless communication is essential for safe wheelchair operation.

The firmware continuously monitors the Bluetooth connection status.

If communication with the Flutter application is interrupted unexpectedly, the ESP32 immediately stops both motors before restarting Bluetooth advertising.

This protection mechanism prevents the wheelchair from continuing to move without user supervision.

Once the application reconnects successfully, normal operation resumes.

5.9.5 Battery Monitoring Algorithm

The firmware continuously estimates the remaining battery level during wheelchair operation.

The battery percentage is transmitted to the mobile application together with the other sensor measurements.

When the battery level falls below the predefined warning threshold, the application automatically displays a low-battery notification to alert the user before the batteries become fully discharged.

This feature reduces the possibility of unexpected wheelchair shutdown during operation.

5.9.6 Temperature Monitoring

The MLX90614 infrared temperature sensor continuously measures the user's body temperature.

If the measured temperature exceeds the normal operating range, the firmware includes the updated value in the transmitted BLE packet.

The Flutter application automatically generates a warning notification to inform the user or caregiver that medical attention may be required.

Continuous temperature monitoring adds an additional healthcare function to the wheelchair.

Engineering Insight

The implemented safety algorithms operate independently of the user's commands. Instead of waiting for user intervention, the embedded controller continuously evaluates system conditions and automatically performs protective actions whenever necessary.

This proactive approach improves reliability and significantly reduces the possibility of accidents.

Design Challenge

One of the main challenges during software development was ensuring that safety algorithms responded immediately without affecting normal wheelchair operation.

This challenge was addressed by continuously monitoring safety conditions inside the firmware while keeping computationally intensive medical processing in a separate FreeRTOS task.

Practical Note

All implemented safety functions were verified during practical testing.

Obstacle detection successfully stopped the wheelchair before collision, fall detection generated emergency notifications, Bluetooth disconnection immediately disabled motor movement, and battery monitoring correctly displayed warning notifications inside the mobile application.

5.10 Software Testing

After completing the firmware development, a series of software tests were performed to verify the reliability and stability of the BioMove embedded system. These tests focused on validating communication, sensor acquisition, motor control, safety functions, and overall system responsiveness.

Each software module was initially tested independently before conducting full system integration tests.

5.10.1 Bluetooth Communication Test

The Bluetooth Low Energy (BLE) communication was tested by establishing a connection between the ESP32 and the Flutter mobile application.

The test verified that:

- The application successfully detected the ESP32.
- A stable BLE connection was established.
- Movement commands were transmitted correctly.
- Sensor data was received every second.
- Automatic reconnection was possible after disconnection.

The communication remained stable throughout continuous operation.

5.10.2 Sensor Verification Test

Each sensor was tested individually before integrating the complete system.

The following measurements were verified:

Sensor	Tested Function
MAX30102	Heart Rate and SpO ₂
MLX90614	Body Temperature
MPU6050	Tilt Detection
Ultrasonic Sensor	Obstacle Distance
NEO-6M GPS	Real-Time Location
Battery Monitor	Battery Percentage

The measured values were successfully displayed on the mobile application.

5.10.3 Motor Control Test

The motor control algorithm was tested under all supported movement commands.

The wheelchair successfully performed:

- Forward movement.
- Reverse movement.
- Left turning.
- Right turning.
- Immediate stop.

Motor speed adjustment through the Flutter application was also verified.

5.10.4 Safety Function Test

Several safety scenarios were intentionally created during testing.

The firmware successfully:

- Stopped the wheelchair when an obstacle was detected.
- Generated a FALL notification after abnormal tilt detection.
- Displayed battery warning notifications.
- Displayed temperature warning notifications.
- Prevented unsafe movement commands.

These tests confirmed that the implemented safety algorithms operated correctly.

5.10.5 Communication Loss Test

To verify system reliability, the Bluetooth connection was intentionally interrupted while the wheelchair was moving.

Immediately after detecting the communication loss, the ESP32:

- Stopped both motors.
- Cancelled the current movement command.
- Restarted BLE advertising.
- Waited for a new connection.

This behavior confirmed the effectiveness of the communication loss protection mechanism.

5.10.6 Continuous Operation Test

The complete embedded system was operated continuously while monitoring all software functions.

During testing:

- Sensor readings remained stable.
- The Flutter application received updates every second.
- Motor response remained smooth.
- GPS coordinates updated correctly.
- Safety algorithms continued operating normally.

No unexpected software crashes or communication failures were observed.

Engineering Insight

Testing individual software modules before full system integration simplified debugging and reduced development time. This modular testing approach made it easier to identify and resolve software issues during implementation.

Practical Note

The final integrated firmware demonstrated stable performance during practical experiments. The embedded software successfully coordinated sensing, communication, motor control, and safety monitoring while maintaining reliable real-time operation.

Test	Expected Result	Actual Result	Status
BLE Communication	Successful Connection	Passed	✓
Sensor Reading	Correct Measurements	Passed	✓
Motor Control	Correct Movement	Passed	✓
Obstacle Detection	Automatic Stop	Passed	✓
Fall Detection	Warning Notification	Passed	✓
Battery Monitoring	Low Battery Alert	Passed	✓
Temperature Monitoring	High Temperature Alert	Passed	✓
GPS Tracking	Correct Map Position	Passed	✓
Communication Loss	Motors Stop Automatically	Passed	✓
Continuous Operation	Stable Performance	Passed	✓

CHAPTER SIX

System Implementation and Experimental Results

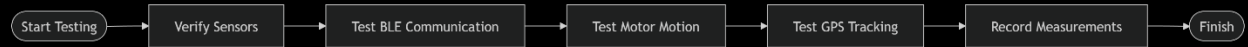
6.1 Introduction

After completing the hardware design and embedded software development, the BioMove system was assembled and evaluated through a series of practical experiments. The objective of these experiments was to verify that all hardware components, embedded firmware, and the Flutter mobile application operated together as an integrated intelligent wheelchair.

The implementation phase involved mechanical assembly, electrical wiring, embedded firmware deployment, mobile application testing, and system validation under different operating conditions.

Experimental testing focused on evaluating the performance of the wheelchair, the responsiveness of the control system, the reliability of wireless communication, and the effectiveness of the implemented safety algorithms.

The results presented in this chapter demonstrate the practical capabilities of the BioMove system and validate the engineering decisions made during the design process.



6.2 Prototype Implementation

The BioMove prototype was constructed as a full-scale smart medical wheelchair rather than a laboratory-scale demonstration model.

A custom steel chassis was designed and fabricated to provide sufficient structural strength while supporting all electronic components, batteries, motors, and the seated user.

The wheelchair employs a six-wheel configuration. The two large middle wheels serve as the primary driving wheels and are powered independently by two 24 V, 250 W DC wheelchair motors. The remaining wheels improve stability and distribute the total load more effectively.

The complete electronic system, including the ESP32 controller, motor drivers, sensors, Bluetooth module, GPS receiver, and power distribution components, was securely mounted on the chassis to ensure reliable operation during movement.

Power is supplied by two 12 V, 12 Ah rechargeable batteries connected in series to provide a 24 V supply for the entire system.

The final prototype integrates mechanical design, embedded electronics, wireless communication, and healthcare monitoring into a single intelligent mobility platform.

6.3 Experimental Setup

A series of practical experiments was conducted to evaluate the overall performance of the BioMove smart medical wheelchair.

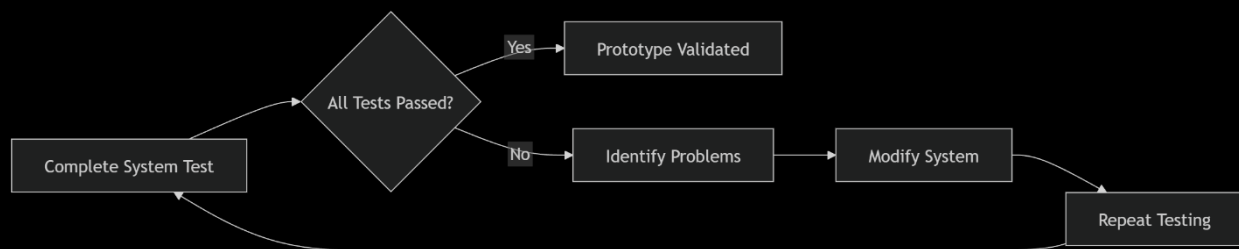
The experiments were performed under normal operating conditions using the completed prototype and the Flutter mobile application.

The testing procedure included:

- Bluetooth communication testing.
- Motor performance evaluation.
- Sensor verification.
- GPS tracking validation.
- Medical monitoring verification.
- Safety algorithm evaluation.
- Battery monitoring verification.
- Real-time application testing.

All experiments were repeated several times to ensure consistent system behavior and reliable operation.

Test	Result
Maximum Load	100 kg
Maximum Speed	8 km/h
Battery Voltage	24 V
Battery Capacity	24 V (2 × 12 V, 12 Ah)
Application Update Rate	1 second
Bluetooth Communication	Stable
GPS Tracking	Successful
Obstacle Detection	Successful
Automatic Stop	Successful
Fall Detection	Successful



EMG Sensor Test

The EMG sensor was tested to verify its ability to detect muscle activity and transmit the measured data to the mobile application in real time. Surface electrodes were attached to the user's forearm, and voluntary muscle contractions were performed under different conditions.

During each contraction, the ESP32 successfully acquired the analog EMG signal through its built-in ADC, processed the acquired values, and transmitted the data to the Flutter application via Bluetooth Low Energy (BLE). The displayed muscle activity level responded immediately to changes in muscle contraction intensity, demonstrating stable communication and reliable signal acquisition.

The experimental results confirmed that the EMG monitoring system operated correctly and can be used as an effective rehabilitation monitoring feature while providing a foundation for future muscle-controlled wheelchair applications.

CHAPTER SEVEN

Conclusion and Future Work

7.1 Conclusion

This project presented the design and implementation of **BioMove**, a smart medical wheelchair that integrates embedded systems, mobile application development, and healthcare monitoring into a single assistive mobility platform.

The system was built around an ESP32 microcontroller and combines two independent DC wheelchair motors, multiple medical and environmental sensors, Bluetooth Low Energy communication, GPS tracking, and a Flutter-based mobile application.

The developed wheelchair provides real-time monitoring of heart rate, blood oxygen saturation, body temperature, battery level, obstacle distance, wheelchair inclination, and geographical location. These measurements are transmitted wirelessly to the mobile application, where they are displayed through an intuitive user interface.

In addition to monitoring, BioMove incorporates several intelligent safety mechanisms. The embedded firmware continuously evaluates environmental and operational conditions to detect obstacles, abnormal wheelchair inclination, communication loss, and low battery conditions.

Whenever a hazardous situation is detected, the system automatically performs the appropriate protective action, improving user safety and system reliability.

Experimental evaluation demonstrated that the prototype successfully supported a theoretical maximum load of **100 kg**, reached a maximum speed of approximately **8 km/h**, maintained stable Bluetooth communication, and updated sensor information every second. The integrated Flutter application also provided reliable wireless control and real-time visualization of all monitored parameters.

Overall, the project demonstrates that combining embedded systems with mobile technologies can significantly improve the functionality, safety, and usability of powered wheelchairs. The final BioMove prototype achieved the objectives established at the beginning of the project and provides a solid foundation for future research and product development.

7.2 Future Work

Although BioMove successfully achieved its primary objectives, several improvements can be considered in future versions to further enhance system performance and user experience.

Possible future developments include:

- Implementing autonomous navigation using computer vision and artificial intelligence.
- Integrating obstacle avoidance based on cameras or LiDAR sensors.
- Adding Wi-Fi or GSM connectivity to enable remote monitoring over the Internet.
- Developing a cloud-based healthcare platform for long-term patient monitoring.
- Implementing voice control to improve accessibility for users with limited hand movement.
- Increasing battery capacity to extend operating time.
- Improving energy management techniques to reduce power consumption.
- Adding electronic seat adjustment and posture assistance features.
- Integrating electronic health record (EHR) support for medical institutions.
- Developing predictive maintenance algorithms to monitor motor and battery health.

These improvements would transform BioMove from a smart powered wheelchair into a comprehensive intelligent healthcare and mobility platform capable of supporting a wider range of users and medical applications.

Final Remark

The development of BioMove represents more than the implementation of an engineering project; it reflects the integration of mechanical design, embedded systems, electronics, mobile application development, and healthcare technologies to address a real-world challenge.

Throughout this project, emphasis was placed on designing a reliable, safe, and practical mobility solution capable of improving the quality of life for elderly individuals and people with mobility impairments.

The knowledge and experience gained during the design, implementation, testing, and evaluation of BioMove have provided valuable engineering skills and established a strong foundation for future work in intelligent medical devices and assistive technologies.